

BRIDGEPULSE · BUILD PLAN

BridgePulse prototype build plan

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Approved follow-ups carried out: pad eye, wire lanyard and girder clamp added (new section 3.11, joint 10, step 9; steps renumbered); plate holes, overview, bought parts, bench offcut length and cost updated; all pictures redrawn from the model
0.3	2026-10-02	Amish Chadha	Decisions of 2026-10-02: wiring follows the adopted FieldNode pinout; safety stop S8 names the lanyard's girder clamp and crimped banding
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; cost line reworded
0.1	2026-09-30	Amish Chadha	First build plan; design made constructable (BRP-DDR-004)

BridgePulse prototype build plan

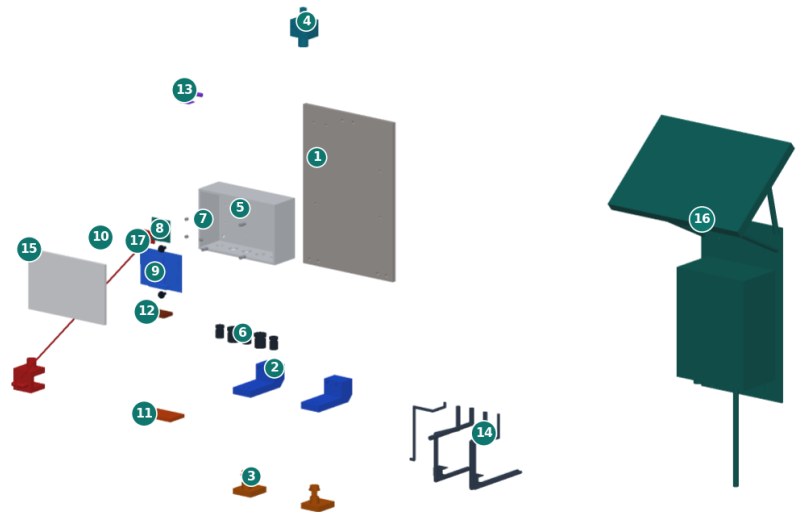
Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register ([docs/06-design-decisions.md](#)), not here.

■ 1. What you are building

BridgePulse prototype: every component, pulled apart

Numbered in build order, seen from the front right and above. The FieldNode core (16) goes on the post; the lanyard (17) runs to its own girder clamp

- 1 ■ Mounting plate
- 2 ■ Foot blocks (2) and M6 screws
- 3 ■ Jaws, packers, M10 bolts (2 sets)
- 4 ■ Jack block, jack screw, M6 screws
- 5 ■ Hub enclosure body, drilled
- 6 ■ Glands and panel connector
- 7 ■ Spacers and standoffs
- 8 ■ Accelerometer board
- 9 ■ Signal board with modules
- 10 ■ M5 hub screws, sealing washers (4)
- 11 ■ Active gauges under covers (2)
- 12 ■ Dummy gauge coupons (2)
- 13 ■ Temperature probes (2, one shown) and clip
- 14 ■ Cables, leads, clips (cut short)
- 15 ■ Hub lid
- 16 ■ FieldNode core with long bands
- 17 ■ Pad eye, wire lanyard, girder clamp



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github.com/BoujeeEnjinia1701/bridgepulse

Figure 1. Every component pulled apart and numbered in build order. The FieldNode core (16) is drawn beside the hub; it goes on the handrail post. The lanyard (17) runs from a pad eye on the plate to its own girder clamp.

The prototype is one BridgePulse monitor fitted to a bench rig that stands in for the middle of the example footbridge: two short offcuts of the girder section, 1.2 m apart, and a stub of the 50 mm square handrail post. At its heart is a sealed die-cast box, the sensor hub, holding an accelerometer board and a signal board. The hub is screwed to an aluminium plate that lies against the girder's web between its flanges, stands on two foot blocks clamped to the bottom flange and is wedged against the top flange by a jack screw, so nothing is drilled into the bridge. A strain gauge under each girder, a dummy gauge on a loose steel coupon beside each, and two temperature probes complete the sensors; cables run to the hub on push-on flange clips, and one cable takes power and data to a FieldNode core on the post. A stainless wire lanyard from a pad eye on the plate to a second girder clamp holds

everything if the mount ever lets go. Figure 1 shows the 17 components in the order you make or fit them. Seven are made in a small workshop: the mounting plate, the foot blocks, the clamp jaws and packers, the jack block, the drilled hub box, the dummy gauge coupons and the temperature probes. Everything else is bought, or in the case of the FieldNode core built to its own plan. The work is sawing, drilling, tapping and filing aluminium and steel bar, drilling a die-cast box, potting two small sensors, bonding strain gauges, and wiring bought modules together. The estimated cost of the BridgePulse parts is about \$275 from the bill of materials (the value-engineering target is \$250), plus the FieldNode core.

Safety: The girder offcuts weigh about 37 kg each: lift them with two people or a hoist and stand them where they cannot tip. Old bridge steel may carry lead paint; test any offcut taken from a bridge before sanding it. Strain gauge adhesives, conditioners and potting epoxy are chemicals: read their safety data sheets, wear gloves and eye protection and ventilate. Cut steel and aluminium edges are sharp: deburr everything. The FieldNode core holds a lithium iron phosphate cell; follow its own plan's safety stops. Fitting the monitor to a real bridge is work at height, outside this plan (section 6, S8).

■ 2. What changed to make it buildable

The concept showed what the monitor does; some of its parts could not be made, fixed or fitted as drawn, chiefly because a rolled girder has rounded root fillets where its web meets its flanges. Each change below keeps what the monitor does, and all of them are recorded in decision record BRP-DDR-004, accepted by Amish on 2026-10-02; the lanyard row follows his decision of the same day.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Mounting plate	A 313 mm plate standing on the bottom flange in the corner against the web	A 290.6 mm plate lying on the web, 22 mm clear of each flange (Figure 5)	The plate would have rested on the 18 mm root fillet, not on the flange or the web
Foot clamps	A loose block, a spacer and a jaw that would tip about the flange edge	A stepped foot block screwed to the plate, a steel jaw under the flange, a packer outside the edge and one bolt (Figures 5 and 7)	Both ends of the jaw bear; nothing but the jaw is below the girder
Jack	A 10 mm tab with no fixing and a screw with no room for its head	A 30 mm jack block screwed to the plate, with the screw's head and lock nut below it (Figure 9)	Enough thread, a spanner from below and a lock
Hub to plate	No fixing	Four screws from inside the box into the plate, with sealing washers (Figure 12)	The box stays sealed and its base is clamped to the plate
Boards in the hub	Signal board on two posts; accelerometer glued	Both boards on four spacers or standoffs (Figure 14)	Firm four-point fixings, clear of the hub screws and gland nuts
Hub penetrations	Two glands and a connector; no way in for the probe leads	Five in one row: two probe glands, two gauge glands, one connector (Figure 13)	Every cable enters through its own seal
Temperature probes	A boss stuck to the web; an air probe hanging from nothing	Steel probe under a flange clip; air probe hanging from the far gauge cable (Figure 19 and step 14)	Fixed without drilling or glue on the bridge
Dummy coupons	Over the root fillet, fixing not stated	35 mm out from the web on a bed of silicone (Figure 17)	Same temperature as the girder, none of its strain
Cables	"Clips" with nothing to clip to	Push-on spring-steel flange clips and ties (Figure 20)	No drilling
FieldNode on the post	V-blocks meant for round poles on a square post	V-blocks left off; plate flat on the post with longer bands (Figure 21)	A flat face seats better on a square post
Lanyard	A plain hole in the plate, which a wire could not pass through with the plate flat on the web, and no anchor on the bridge	A stainless pad eye screwed to the plate and a wire lanyard to a second, separate girder clamp on the bottom flange (Figure 22 and step 9)	A second hold on the bridge that shares nothing with the foot clamps

■ 3. Making the components

Make and check each component before the assembly step that needs it. Sizes are in millimetres. "Left" and "right" are as seen standing in front of the hub, looking at its lid. "Front" is the side away from the girder web. Workshop tolerance is 0.5 mm unless a step says otherwise; drawings do not carry tolerances before TRL 4.

3.1 Mounting plate

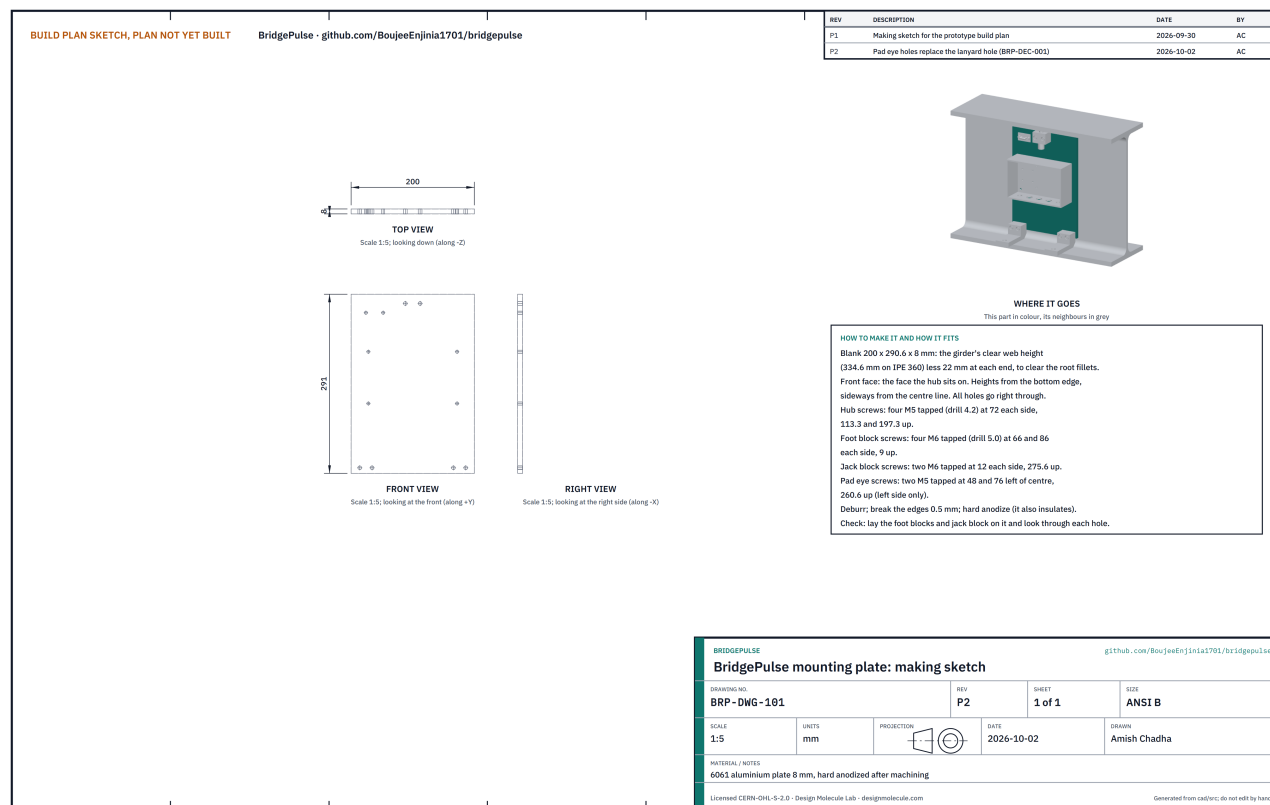
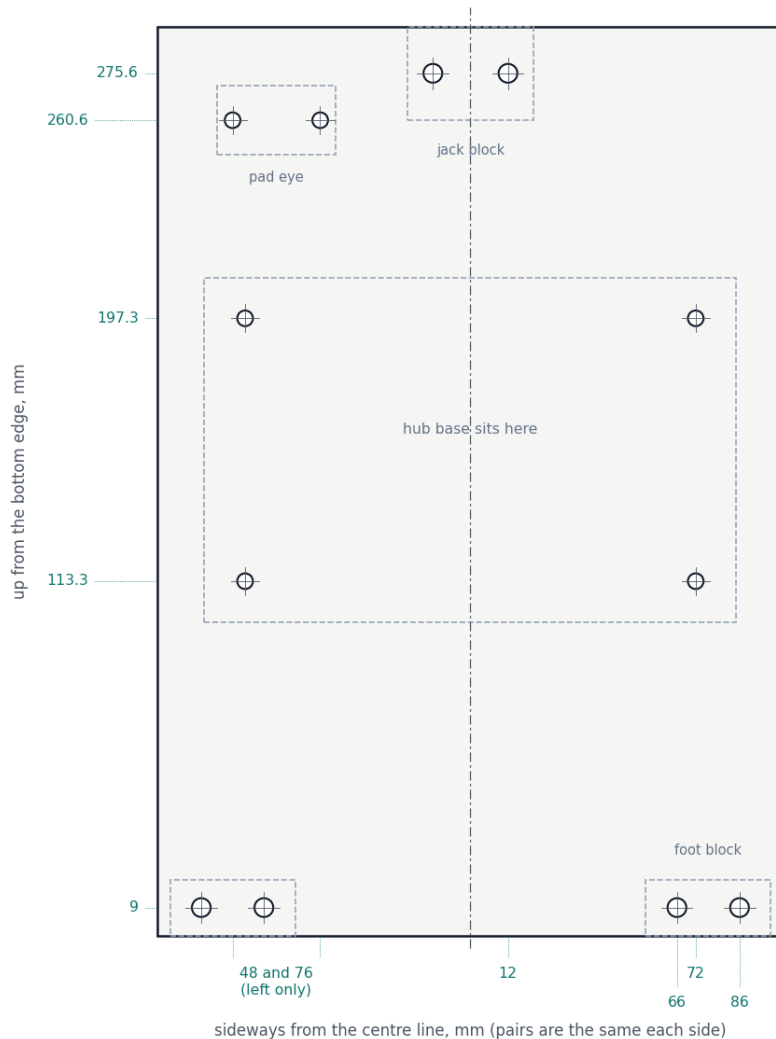


Figure 2. Mounting plate making sketch (BRP-DWG-101).

Mounting plate: hole positions

Seen from the front (the face the hub sits on). Plate 200 x 290.6 x 8 mm. Figures in mm, from the model.



What each hole is

Hub screws: M5 tapped, drill 4.2 (4 holes)

Foot block screws: M6 tapped, drill 5.0 (4)

Jack block screws: M6 tapped, drill 5.0 (2)

Pad eye screws: M5 tapped, drill 4.2 (2), left side only

Every hole goes right through. The back face lies flat on the girder web: no screw may stand out behind it.

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Figure 3. Every hole, measured up from the bottom edge and sideways from the centre line, with the hub, foot blocks, jack block and pad eye outlined.

What it is and what it is made from. The plate everything at the girder hangs on: the hub on its front, the foot blocks at its bottom, the jack block at its top. Its back lies flat on the girder's web. Aluminium plate 8 mm thick, 6061 class, 200 wide; its length is the girder's clear web height (334.6 on an IPE 360) less 22 at each end, 290.6 on the example bridge.

How to make it.

1. Measure the clear height between the flanges of the girder it will go on, next to the web. Cut the blank 200 wide and that height less 44 long, square. File the edges and break them 0.5.
2. Scribe a centre line down the long side. Choose one face as the front and mark it.
3. Mark every hole from Figure 3: heights up from the bottom edge, sideways from the centre line.
4. Hub screws: four holes 72 each side of centre, 113.3 and 197.3 up. Drill 4.2 and tap M5 right through.
5. Foot block screws: four holes 66 and 86 each side of centre, 9 up. Drill 5.0 and tap M6 right through.
6. Jack block screws: two holes 12 each side of centre, 275.6 up. Drill 5.0 and tap M6 right through.
7. Pad eye screws: two holes 48 and 76 left of centre, 260.6 up. Drill 4.2 and tap M5 right through.
8. Deburr every hole on both faces; the back must lie flat on the web, so take off every burr there.
9. Have the plate hard anodized. The anodizing protects it and insulates it from the steel.

How it fits the parts next to it. The back lies flat on the girder web, with 22 mm between each end and the flange, which clears the rounded root fillets (Figure 5). The foot blocks sit flat on the front at the bottom, the hub's base in the middle from 100.3 to 210.3 up, the jack block at the top and the lanyard's pad eye at the top left. No screw may stand out of the back.

Check before moving on. Lay the foot blocks and jack block on it and look through each hole: the holes line up without forcing a screw. Hold the plate against the web of the girder offcut: it fits between the fillets with no rock.

3.2 Foot blocks (make 2)

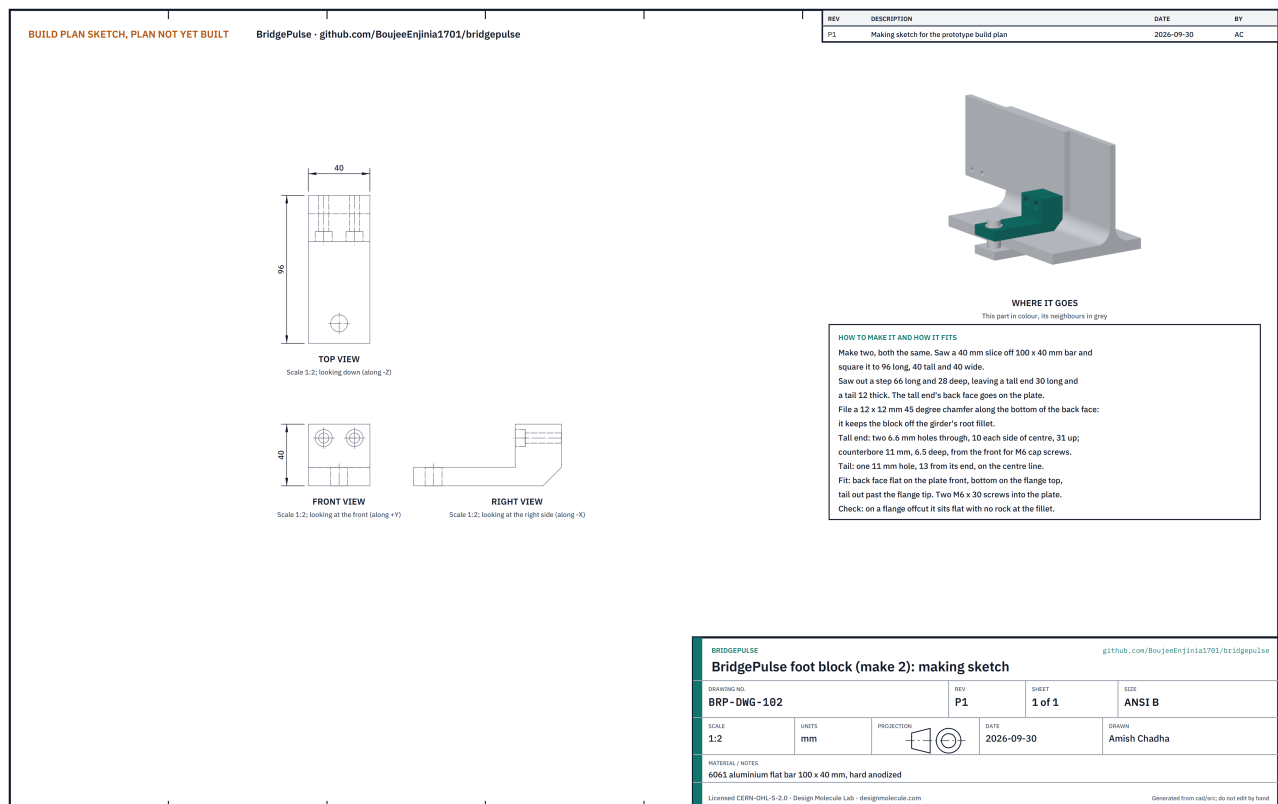


Figure 4. Foot block making sketch (BRP-DWG-102).

What it is and what it is made from. A stepped block that the plate stands on: its tall end is screwed to the plate, its tail lies on the bottom flange and is clamped at the flange edge. Aluminium flat bar 100 x 40, 6061 class.

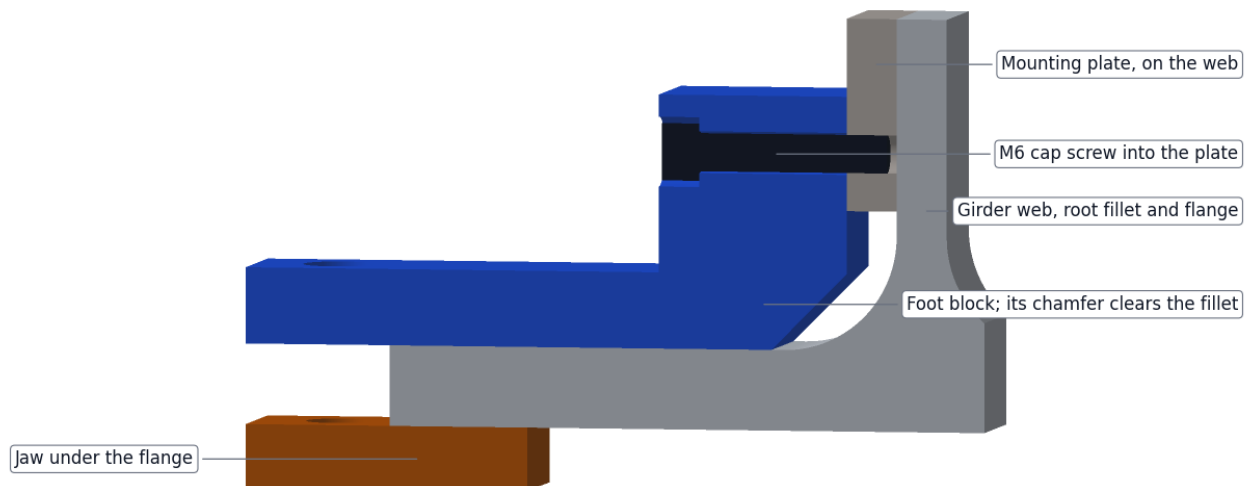
How to make it.

1. Saw a 40 slice off the bar and square it to 96 long, 40 tall and 40 wide.
2. Saw out a step 66 long and 28 deep, leaving a tall end 30 long and a tail 12 thick. File the sawn faces flat. The outside face of the tall end is the back face; it goes on the plate.
3. File a 12 x 12 chamfer at 45° along the bottom edge of the back face. This keeps the block off the girder's root fillet.
4. Tall end: drill two 6.6 holes through, front to back, 10 each side of centre and 31 up from the bottom. Counterbore each 11 across and 6.5 deep from the front for an M6 cap screw.
5. Tail: drill one 11 hole, top to bottom, 13 from its end, on the centre line.
6. Deburr and have the blocks hard anodized with the plate.

How it fits the parts next to it.

Joint 1: foot block, plate and bottom flange (cut through an M6 screw)

Seen along the span. The plate stops 22 mm above the flange; the block's chamfer clears the root fillet



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Figure 5. Cut through an M6 screw: the plate stops 22 mm above the flange, and the block's chamfer clears the root fillet.

The back face sits flat on the plate's front, its top 18 above the plate's bottom edge, and is held by two M6 x 30 cap screws into the plate's tapped holes; the screws end 1.5 short of the web. The block's bottom lies on the flange top and its tail runs 23 past the flange edge, where the clamp holds it (section 3.3).

Check before moving on. On the girder offcut, with the plate against the web, each block sits flat on the flange with no rock at the fillet.

3.3 Clamp jaws and packers (make 2 of each)

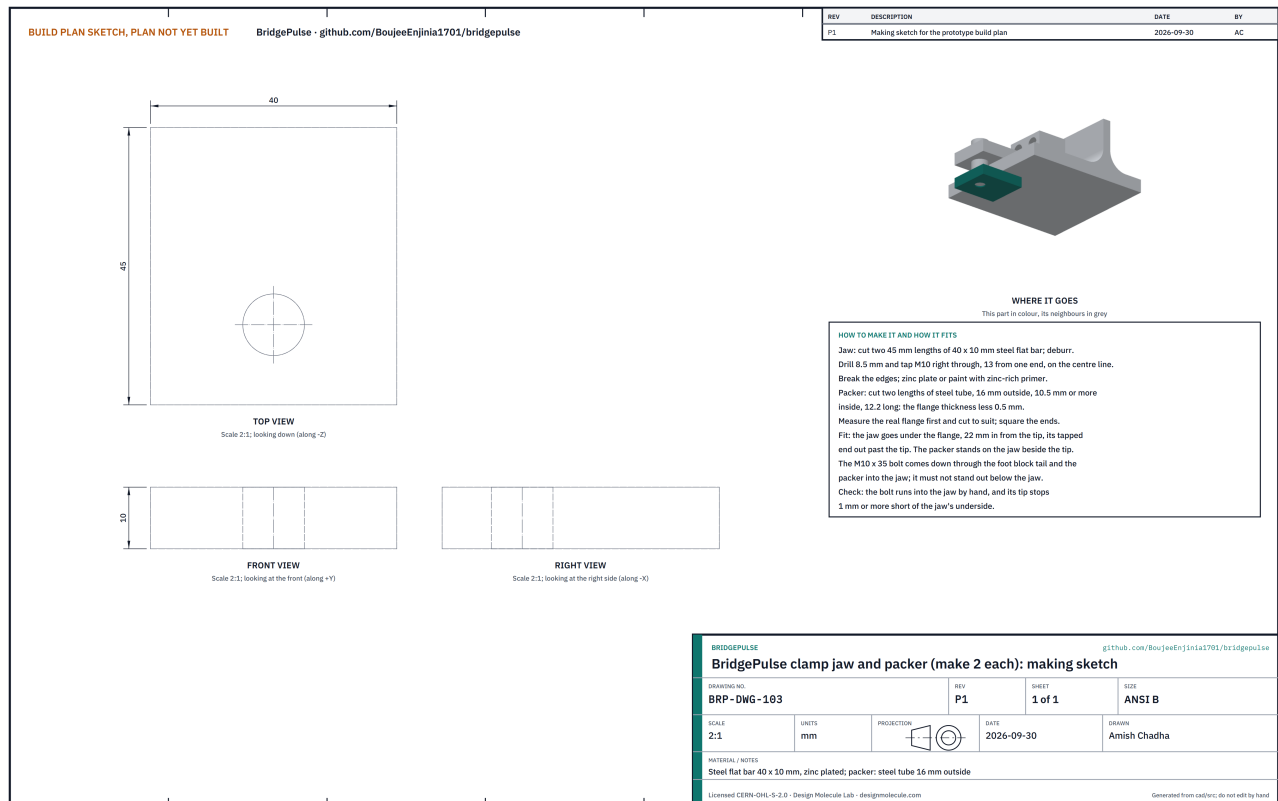


Figure 6. Clamp jaw and packer making sketch (BRP-DWG-103).

What it is and what it is made from. The jaw goes under the bottom flange and the packer stands beside the flange edge; one bolt pulls the jaw up and the foot block's tail down, gripping the flange between them. Jaw: steel flat bar 40 x 10, zinc plated. Packer: steel tube 16 outside, 10.5 or more inside.

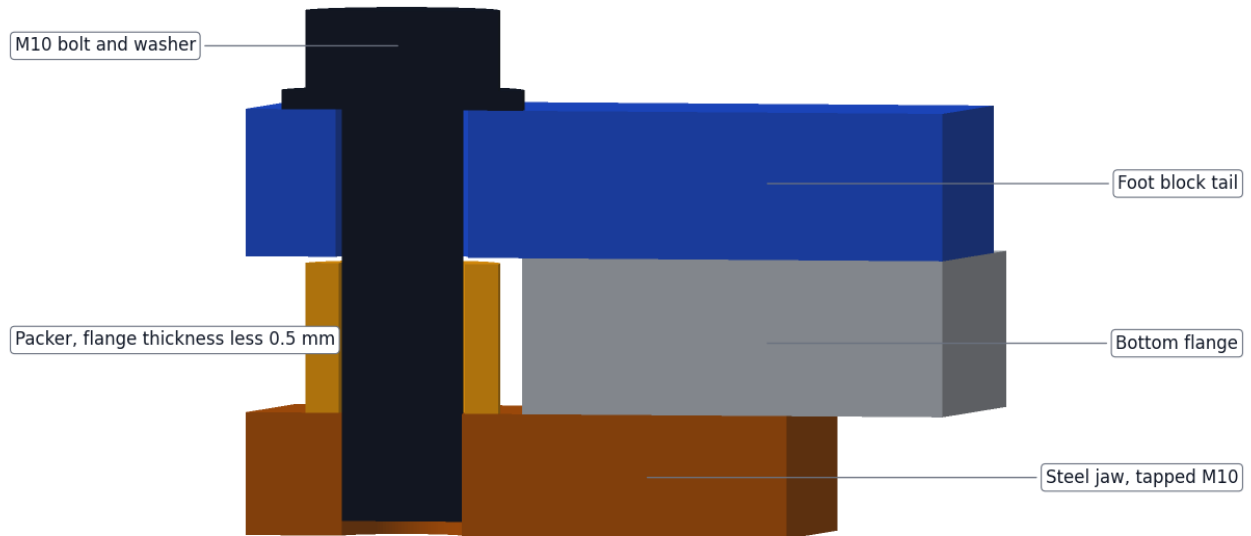
How to make it.

1. Jaw: cut two 45 lengths of the flat bar. Drill 8.5 and tap M10 right through, 13 from one end, on the centre line. Break the edges and zinc plate them, or paint them with zinc-rich primer.
2. Packer: measure the real flange thickness at the flange edge (12.7 on an IPE 360). Cut two lengths of tube 0.5 shorter than that, 12.2 on the example. Square the ends.

How it fits the parts next to it.

Joint 2: the clamp at the flange tip (cut through the bolt)

Tightening the bolt pulls the jaw up under the flange and the tail down on top; nothing below the jaw



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Figure 7. Cut through the bolt: the jaw is under the flange, the packer stands on it outside the flange edge, and the bolt ends inside the jaw.

The jaw lies under the flange, reaching 22 in from the edge, with its tapped end out past the edge. The packer stands on the jaw around the bolt, 2 clear of the flange edge. An M10 x 35 bolt with a washer comes down through the foot block's tail and the packer into the jaw. Because the packer is 0.5 shorter than the flange, tightening the bolt clamps the jaw to the flange's underside and the tail to its top. The bolt's end stops 1.2 short of the jaw's underside, so nothing but the 10 jaw is below the girder.

Check before moving on. The bolt runs into the jaw by hand, and with the jaw on a flat surface the bolt's end does not stand out below it.

3.4 Jack block

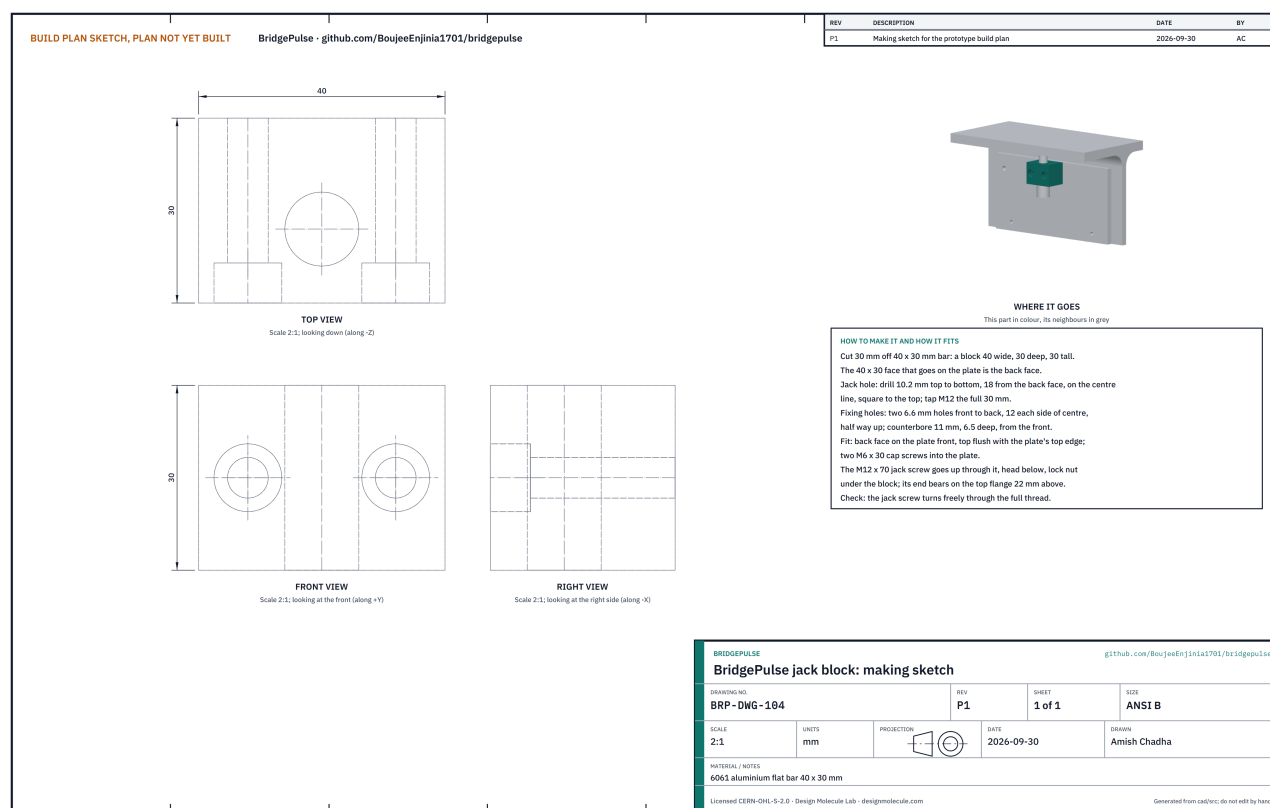


Figure 8. Jack block making sketch (BRP-DWG-104).

What it is and what it is made from. The block at the top of the plate that carries the jack screw. Aluminium flat bar 40 x 30, 6061 class.

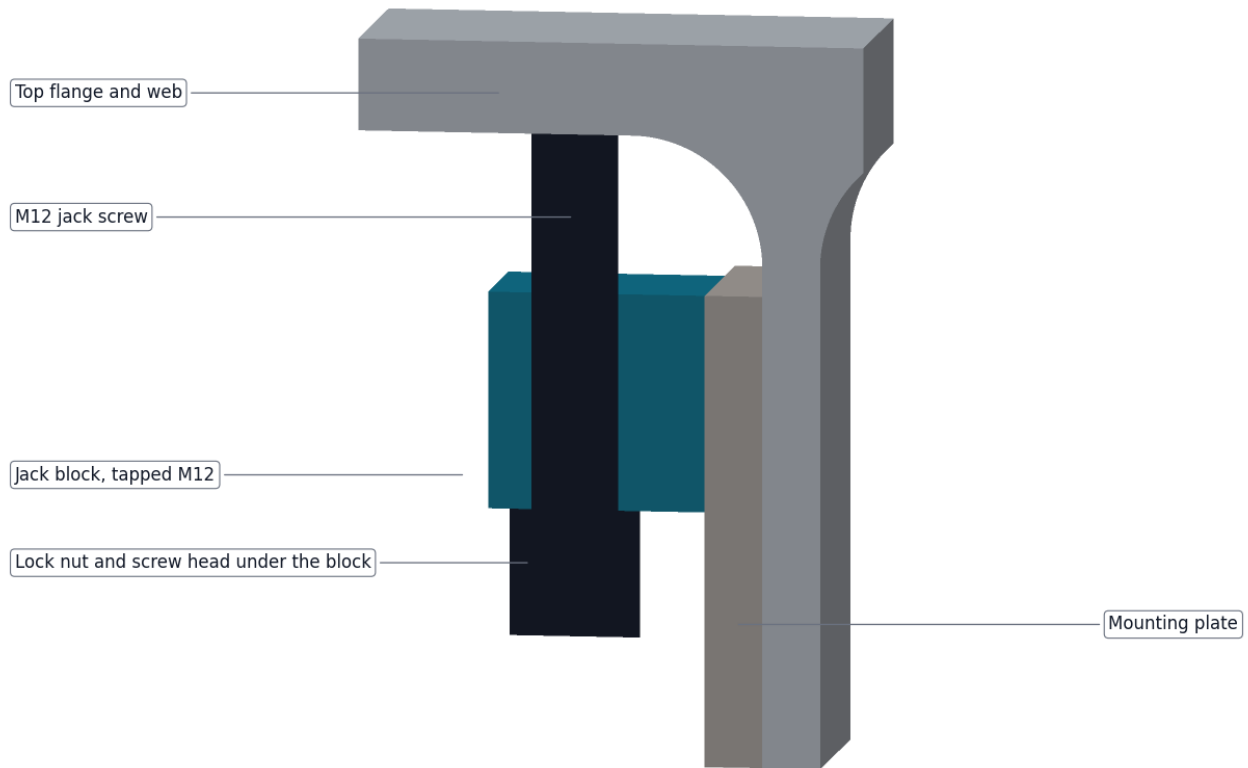
How to make it.

1. Cut 30 off the bar: a block 40 wide, 30 deep and 30 tall. Choose a 40 x 30 face as the back face; it goes on the plate.
2. Jack hole: drill 10.2 top to bottom, 18 from the back face, on the centre line, square to the top. Tap M12 the full 30.
3. Fixing holes: two 6.6 holes front to back, 12 each side of centre and half way up. Counterbore each 11 across and 6.5 deep from the front.
4. Deburr.

How it fits the parts next to it.

Joint 3: jack block and jack screw under the top flange (cut through the screw)

The screw bears on the flange 26 mm out from the web, clear of the root fillet; two M6 screws hold the block



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Figure 9. Cut through the jack screw: it bears on the top flange 26 mm out from the web, clear of the root fillet.

The back face sits on the plate's front with its top flush with the plate's top edge, held by two M6 x 30 cap screws. An M12 x 70 stainless screw goes up through it, head below, with a lock nut under the block. Its end bears on the underside of the top flange, 22 above the block. Winding it up wedges the plate between the flanges; the lock nut holds it there.

Check before moving on. The jack screw turns freely through the full thread and the lock nut runs down to the block.

3.5 Hub enclosure, drilled, with its glands and connector

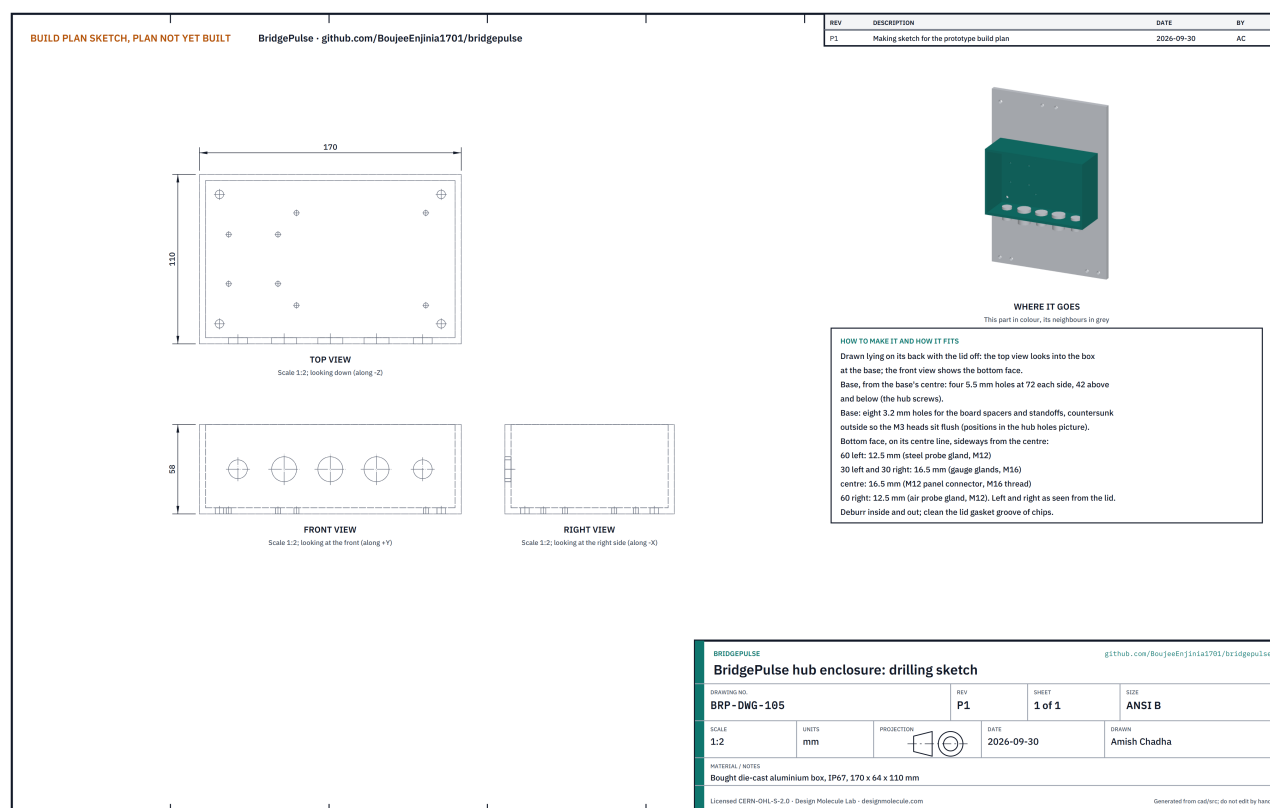
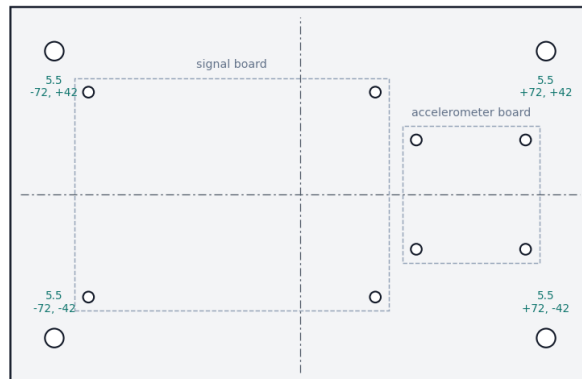


Figure 10. Hub enclosure drilling sketch (BRP-DWG-105), drawn lying on its back.

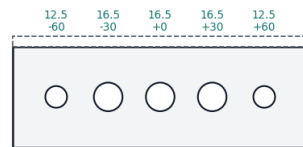
Hub enclosure: holes in the base and the bottom face

Figures in mm from the model, measured from the centre lines of each face.



Base, seen from outside (the face that goes on the plate): left and right are reversed from the lid side; + is to the right in this view.

M3 countersunk holes (3.2): across -62, 22, 34, 66; up -30, -16, +16, +30



Bottom face, seen from below, base edge at the bottom; + to the right as seen from the lid side.
Holes on the face's centre line, 29 from the base.

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Base

4 x 5.5: hub screws to the plate

8 x 3.2, countersunk outside:

board spacers and standoffs

Bottom face (from the lid side)

60 left: steel probe gland, M12

30 left: near gauge gland, M16

centre: M12 panel connector

30 right: far gauge gland, M16

60 right: air probe gland, M12

Seal each countersunk M3 with

thread sealant: the plate covers

the heads once the hub is on.

Figure 11. Holes in the base, seen from outside, and in the bottom face, seen from below.

What it is and what it is made from. A bought die-cast aluminium box, 170 wide, 64 deep and 110 tall, rated IP67, with a gasketed lid. Its base, the large face opposite the lid, is screwed flat to the plate; its bottom face takes the five penetrations.

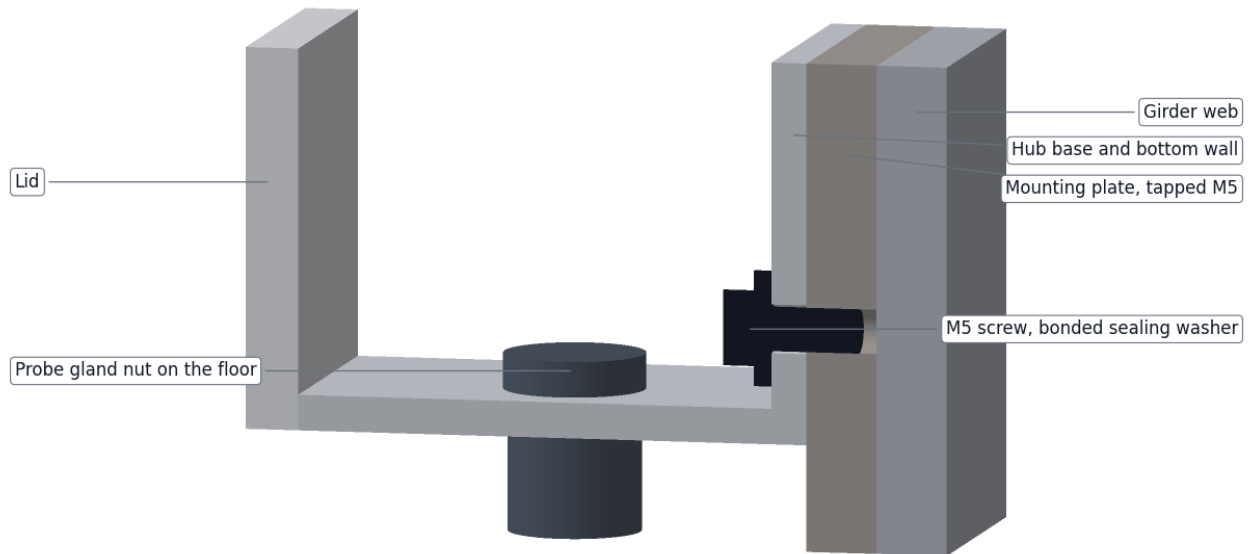
How to make it.

1. Take the lid off and keep its screws. Lay the box on a soft cloth.
2. Base, from Figure 11: four 5.5 holes at 72 each side of the base's centre, 42 above and below it (the hub screws). Eight 3.2 holes for the board spacers and standoffs, countersunk from outside so M3 countersunk heads sit flush. Mind that Figure 11 shows the base from outside, so left and right are reversed.
3. Bottom face, on its centre line, 29 from the base: a 12.5 hole 60 left of centre (steel probe gland), 16.5 holes 30 left and 30 right (gauge glands), a 16.5 hole at the centre (panel connector) and a 12.5 hole 60 right (air probe gland). Pilot drill each 3, then open out with a step drill. Check each size against the part's datasheet before the last step.
4. Deburr inside and out and clean every chip out of the box and the lid's gasket groove.
5. Fit the five penetrations from below, seal outside and nut inside, to the maker's torque: M12 glands at 60 left and right, M16 glands at 30 left and right, the M12 5-pin panel connector in the centre.
6. Fit the eight board spacers and standoffs inside the base on M3 countersunk screws from outside, with a drop of thread sealant on each: four 5 metal spacers for the accelerometer board on the left, four 14 standoffs for the signal board to the right.

How it fits the parts next to it.

Joint 4: hub base on the plate (cut through the lower right hub screw)

The screw goes in from inside the box, sealing washer under its head; its tip stops 2 mm short of the web



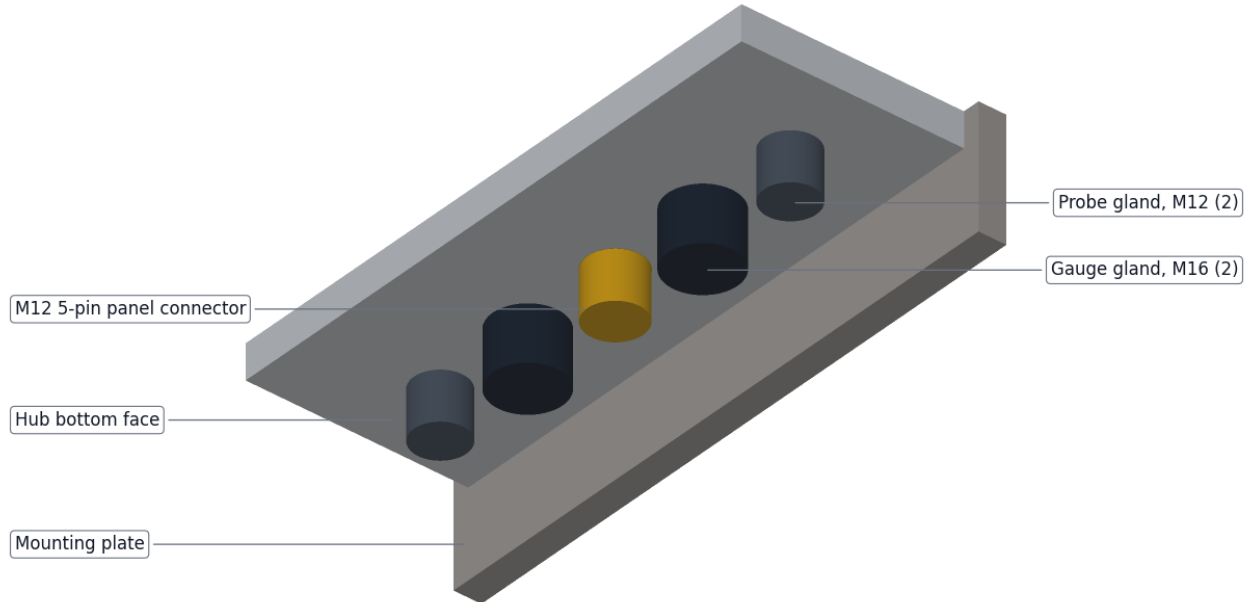
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Figure 12. Cut through a hub screw: it goes in from inside the box, with a bonded sealing washer under its head, and its tip stops 2 mm short of the web.

Joint 5: the hub's bottom face, seen from below and in front

Five penetrations in one row: probe, gauge, connector, gauge, probe; 9 mm or more between them



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Figure 13. The five penetrations, 9 mm or more apart.

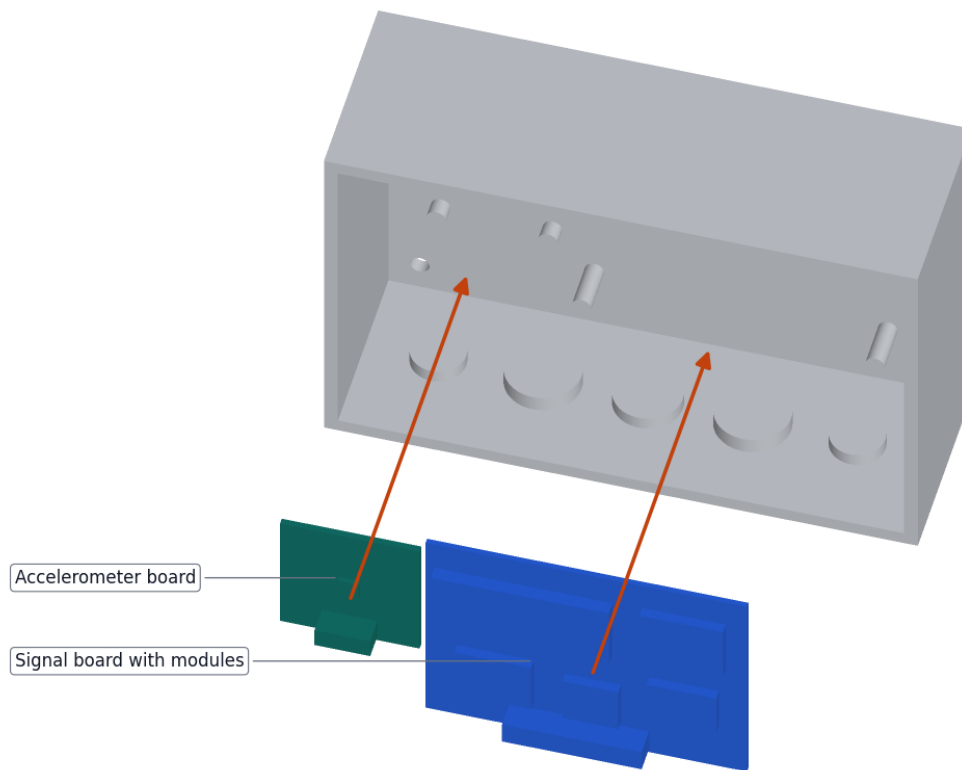
The base lies flat on the plate's front, centred, from 100.3 to 210.3 up the plate. Four M5 x 12 screws go from inside the box through the base into the plate, each with a bonded sealing washer under its head. Once the hub is on, the plate covers the countersunk M3 heads. The front face of the closed hub is 9 inside the line of the flange edge, so the hub sits inside the girder's outline.

Check before moving on. Every penetration seats evenly on its seal; the base sits flat on the plate with all four screw holes lined up; the lid closes on a clean gasket.

3.6 Boards in the hub

Step 4: boards into the hub, then wire them

Accelerometer board on its four spacers, signal board on its standoffs; wire as the wiring diagram shows



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Figure 14. The accelerometer board goes on the four short spacers on the left, the signal board on the four standoffs on the right.

What it is and what it is made from. Bought modules: an ADXL355 class accelerometer board and a signal board made up of an RP2040 class controller board, a 24-bit bridge ADC module (ADS1220 class), a 5 V to 3.3 V buck converter, an RS-485 transceiver module, a microSD socket with a 32 GB card, the bridge completion resistors, an excitation switch and a terminal block, all on a 92 x 68 perforated carrier board.

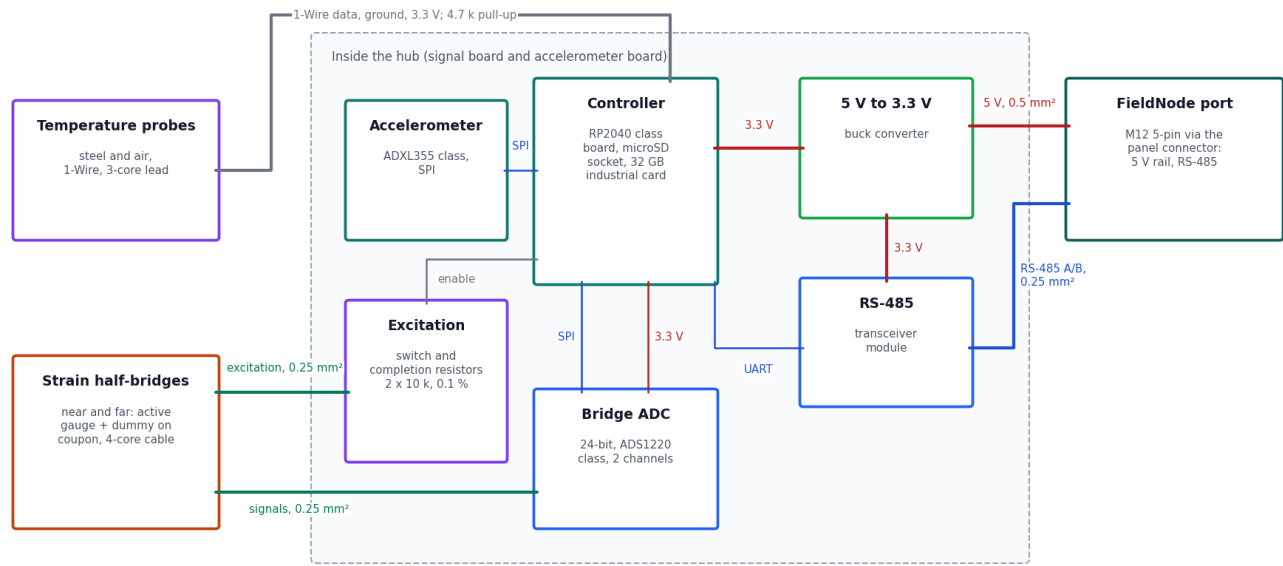
How to make it.

1. Lay the modules out on the carrier board as Figure 14 shows: controller top left, ADC module top right, RS-485 module below it, buck converter lower left, microSD socket lower middle, terminal block along the bottom edge.
2. Drill the carrier's four 3.2 corner holes, 4 in from each edge, to match the standoffs.
3. Fit the modules on short M2.5 or M3 standoffs or solder them in, and wire them as Figure 15 shows (section 3.6.1).
4. Fit the accelerometer board on its four spacers and the carrier on its four standoffs, with M3 screws from inside.

3.6.1 Wiring

BridgePulse prototype: block-level wiring of the sensor hub

Bought modules wired at block level on the signal board; no circuit board is laid out. Stranded copper; ferrules on every screw terminal.



Red: power. Blue: data buses. Green: bridge excitation and signals (shields to ground at the hub end only). Grey: control and 1-Wire.

Extra-low voltage only: 5 V at the port, 3.3 V inside. Gauge leads twisted; probe leads away from the excitation pair.

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Figure 15. Block-level wiring with wire sizes. No circuit board is laid out at this stage; bought modules stand in for it.

Wire it like this, with stranded copper and a ferrule on every screw terminal:

1. Panel connector, switched 5 V rail and ground, to the buck converter's input: 0.5 mm² (20 AWG).
2. Buck converter's 3.3 V output to the controller, the ADC module, the RS-485 module and the accelerometer board: 0.5 mm².
3. Panel connector's two RS-485 pins to the RS-485 module's A and B: 0.25 mm² (24 AWG), twisted. The pin assignment follows the FieldNode pinout adopted as the shared standard (pin 1 switched 5 V rail, pins 2 and 4 RS-485 A and B, pin 3 ground, pin 5 analog; Modbus RTU).
4. Controller to the RS-485 module (UART), to the ADC module and to the accelerometer board (SPI): short 0.25 mm² leads.
5. Controller output to the excitation switch's enable input: 0.25 mm².
6. Each gauge cable at the terminal block: excitation pair from the excitation switch, signal pair to the ADC module through the completion resistors, shield to ground at the hub end only.
7. Both probe leads at the terminal block on one 1-Wire bus: data to a controller pin with a 4.7 k pull-up to 3.3 V, ground and 3.3 V.

Check before moving on. Every wire continues end to end; with nothing connected, the 5 V and 3.3 V rails read open to ground; every wire is labelled.

3.7 Dummy gauge coupons (make 2)

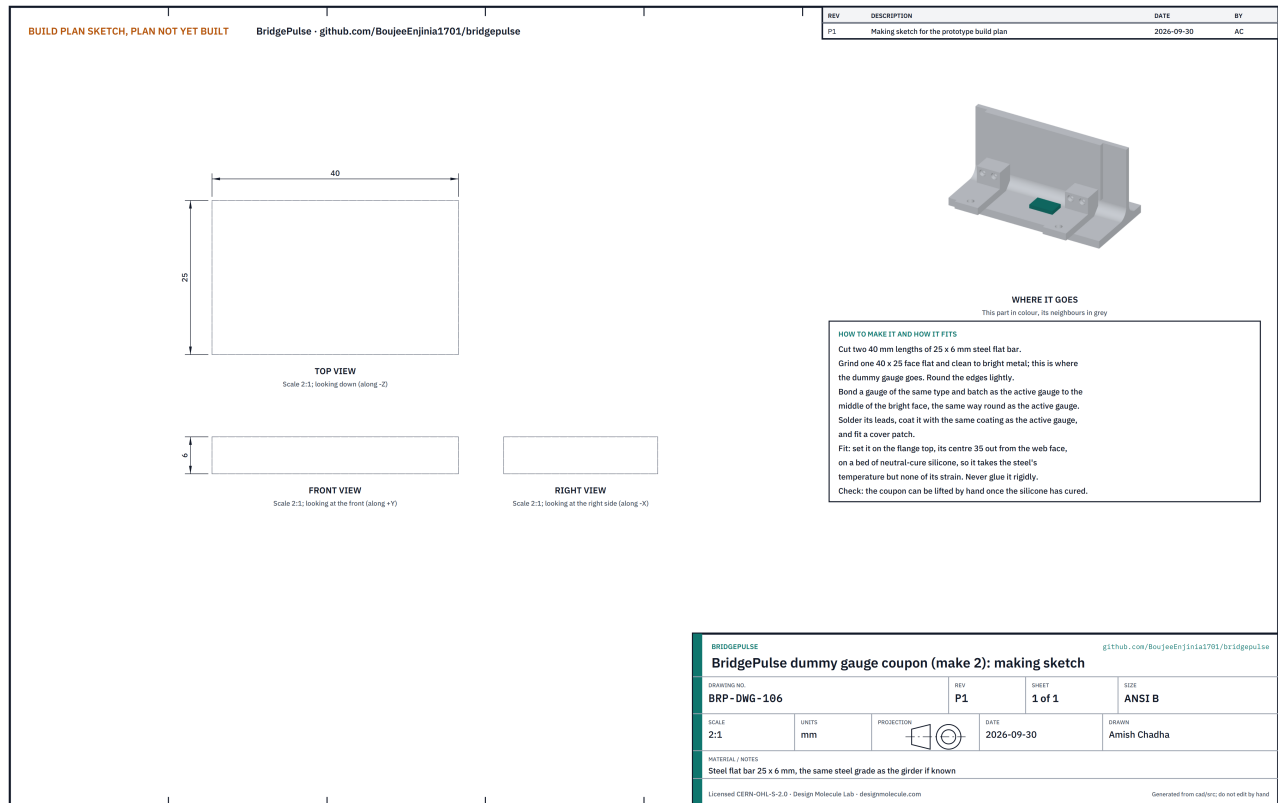


Figure 16. Dummy gauge coupon making sketch (BRP-DWG-106).

What it is and what it is made from. A small loose piece of steel carrying a second gauge that sees the girder's temperature but none of its strain, so the half-bridge cancels temperature. Steel flat bar 25 x 6, the same grade as the girder if known.

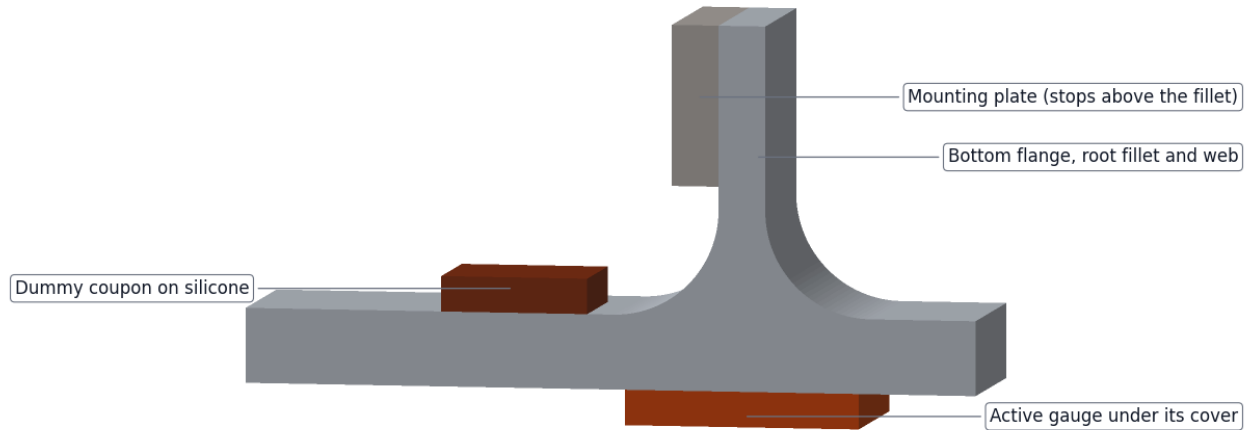
How to make it.

1. Cut two 40 lengths of the flat bar. Round the edges lightly.
2. Grind one 40 x 25 face flat and clean it to bright metal.
3. Bond a gauge of the same type and batch as the active gauge to the middle of that face, the same way round, following the gauge maker's procedure.
4. Solder its leads, coat it with the same coating as the active gauge and fit a cover patch.

How it fits the parts next to it.

Joint 6: active gauge and dummy coupon (cut across the south girder at midspan)

The active gauge is bonded under the flange on the girder's centre line; the coupon only rests on silicone



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github.com/BoujeeEnjinia1701/bridgepulse

Figure 17. Cut across the south girder at midspan: the active gauge under the flange, the coupon on the flange top.

The coupon lies on the top of the bottom flange, its centre 35 out from the web face (clear of the root fillet), on a bed of neutral-cure silicone that holds it in place without passing strain into it. Never glue it rigidly.

Check before moving on. Once the silicone has cured, the coupon can be lifted by hand.

3.8 Temperature probes (make 2)

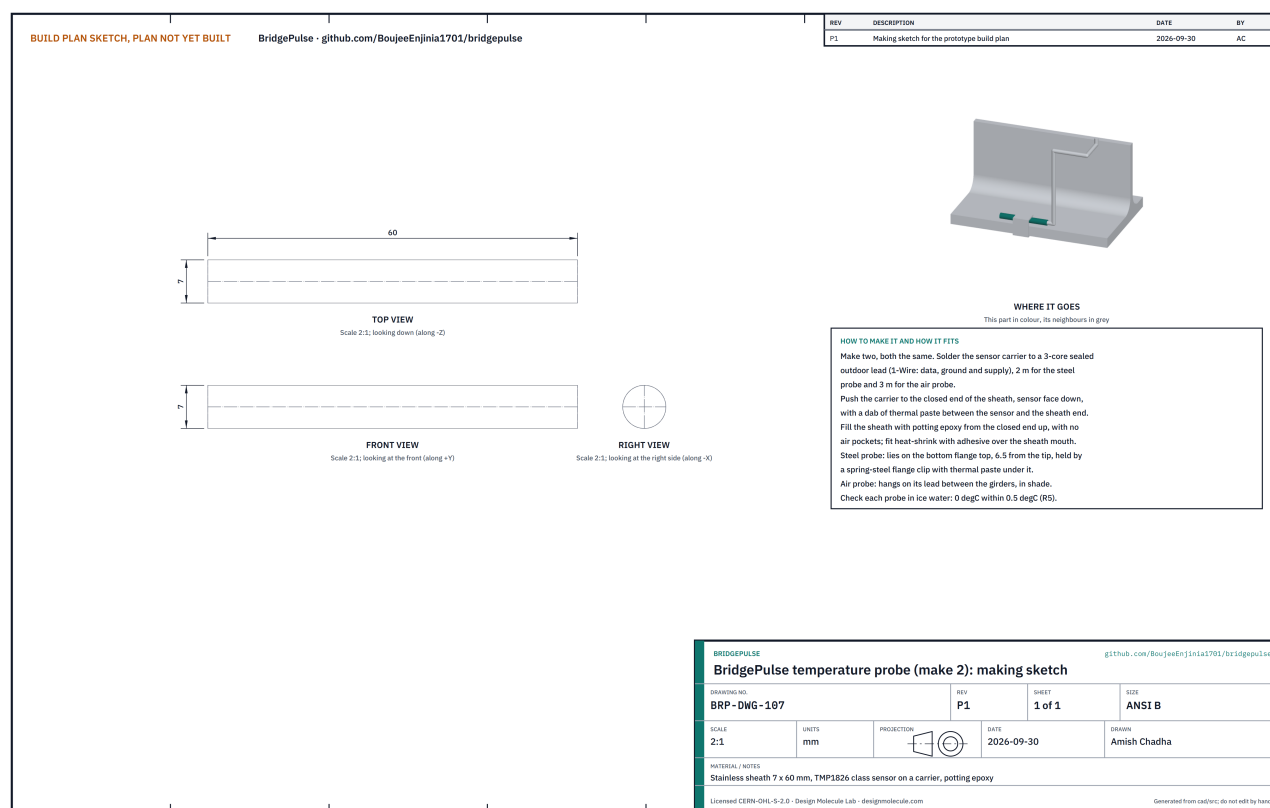


Figure 18. Temperature probe making sketch (BRP-DWG-107).

What it is and what it is made from. A TMP1826 class digital temperature sensor on a small carrier, potted in a stainless sheath 7 across and 60 long, with a sealed outdoor lead. One measures the steel, one the shaded air.

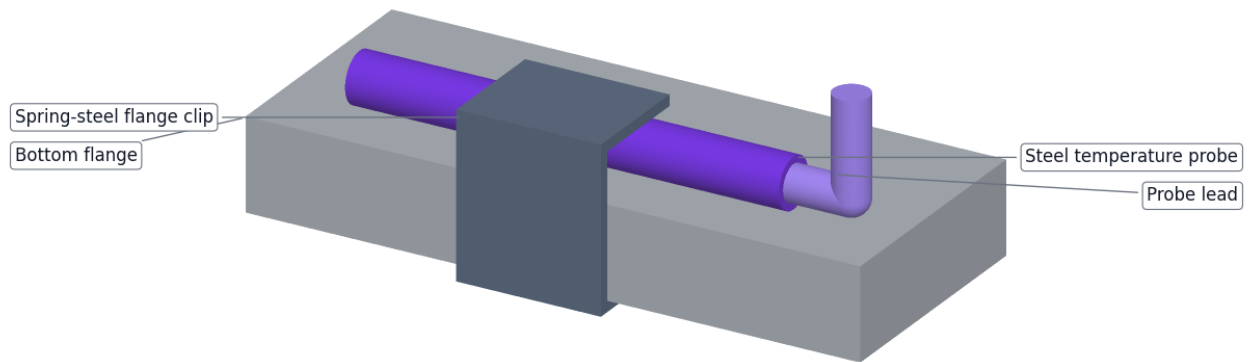
How to make it.

1. Solder the sensor carrier to a 3-core sealed outdoor lead: about 2 m for the steel probe and 3 m for the air probe.
2. Push the carrier to the closed end of the sheath, sensor face down, with a dab of thermal paste between the sensor and the sheath.
3. Fill the sheath with potting epoxy from the closed end up, with no air pockets. When it has cured, fit adhesive-lined heat-shrink over the sheath's mouth and the lead.

How it fits the parts next to it.

Joint 7: steel temperature probe on the bottom flange

Seen from the front, below the deck. The clip pushes the sheath down on a smear of thermal paste



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Figure 19. The steel probe lies on the bottom flange, 6.5 mm in from its edge, under a push-on flange clip.

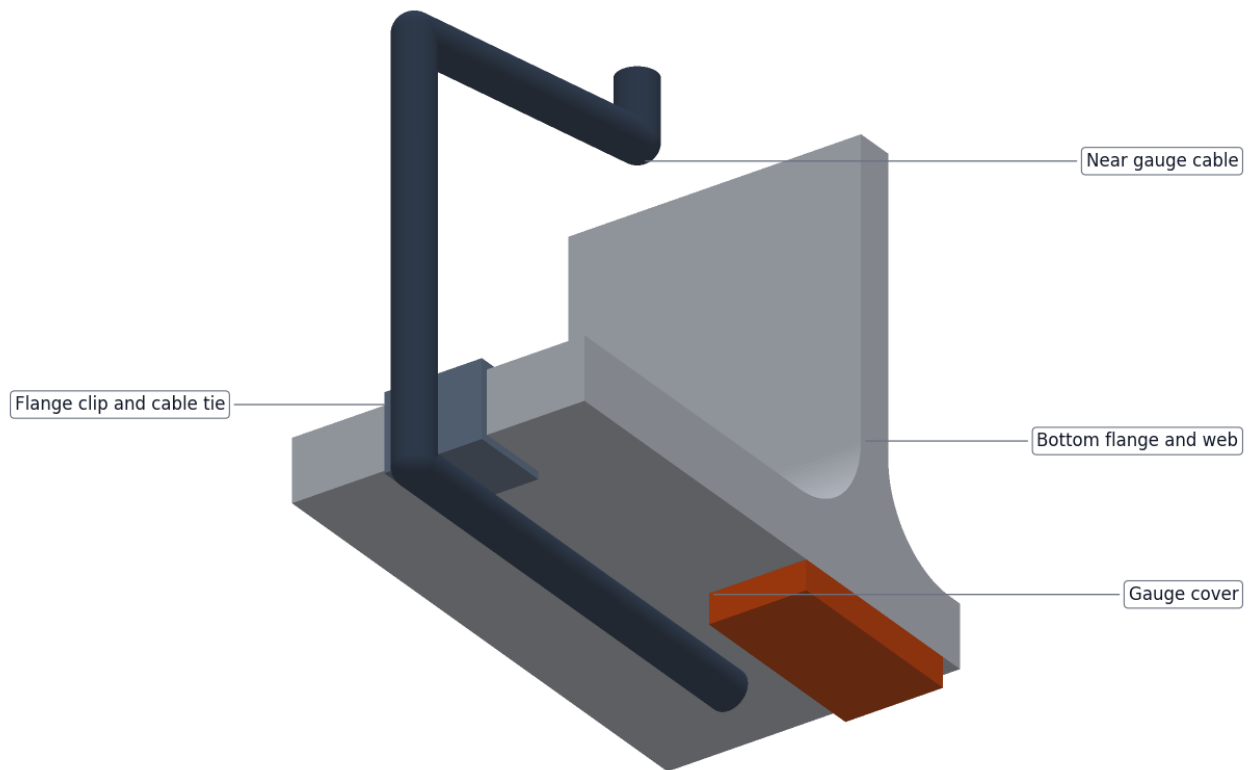
The steel probe lies along the top of the bottom flange, 140 left of centre and 6.5 in from the flange edge, on a smear of thermal paste, held down by a spring-steel flange clip pushed onto the flange edge. Its lead runs up to the hub's left probe gland. The air probe hangs on its lead in the shade between the girders (step 14).

Check before moving on. Put both probes in a bath of crushed ice and water with a reference thermometer: each reads 0 °C within 0.5 °C (R5).

3.9 Cables, clips and the M12 cable (bought)

Joint 8: gauge cable round the flange tip

Seen from below and in front. The cable runs 2 mm under the flange and is tied to a push-on flange clip



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Figure 20. The near gauge cable leaves the gauge cover, runs 2 mm under the flange and round its edge, tied to a push-on flange clip.

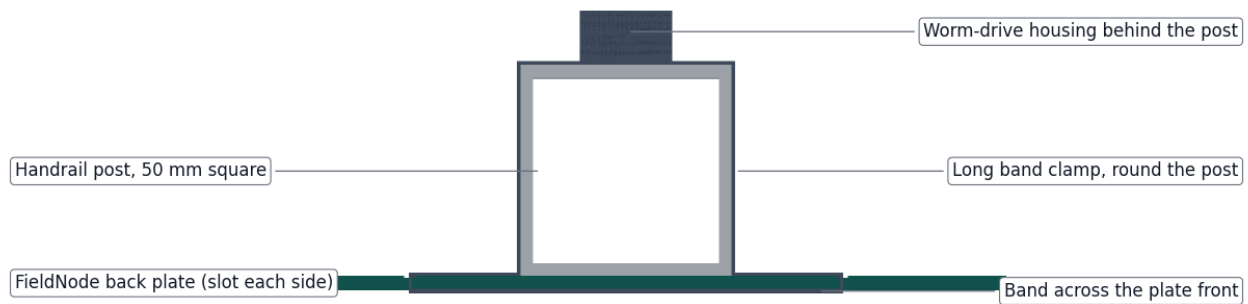
What it is. Two shielded 4-core outdoor gauge cables, one moulded M12 5-pin cable about 2 m long from the hub's panel connector to the FieldNode port, five push-on spring-steel flange clips for 10 to 16 flanges (one of them holds the steel probe) and UV-stable cable ties. Nothing is drilled.

How it fits. Each gauge cable leaves its gauge cover, runs under the bottom flange clipped 2 below it, turns round the flange edge on a flange clip, rises beside the edge and goes into its gland from below: the near cable into the left gauge gland and the far cable into the right one. The far cable crosses between the girders under the deck, 3 inboard of each flange edge. The M12 cable leaves the panel connector, runs out past the flange edge, rises beside the post and plugs into the FieldNode's port B.

3.10 FieldNode core (built to its own plan)

Joint 9: FieldNode on the square post (cut at the lower band, seen from above)

V-blocks left off: the back plate bears flat on the post; the band goes round the post and through both slots



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github.com/BoujeeEnjinia1701/bridgepulse

Figure 21. Cut at the lower band, seen from above: the back plate flat on the post, the band round the post and through both slots.

What it is. The lab's shared power and radio node: a sealed box with a cell, charger, LoRaWAN radio and two M12 sensor ports, a 6 W solar panel on a bracket above it, and a back plate with band clamps. Build it to its own plan, FND-BLD-001, with two differences for a square handrail post:

1. Leave the two V-blocks off. Their screw holes in the back plate stay empty.
2. Buy its two band clamps one size longer, with a band of about 300 to 330 mm, since each band goes round the 50 mm post, through both slots in the back plate and across its front.

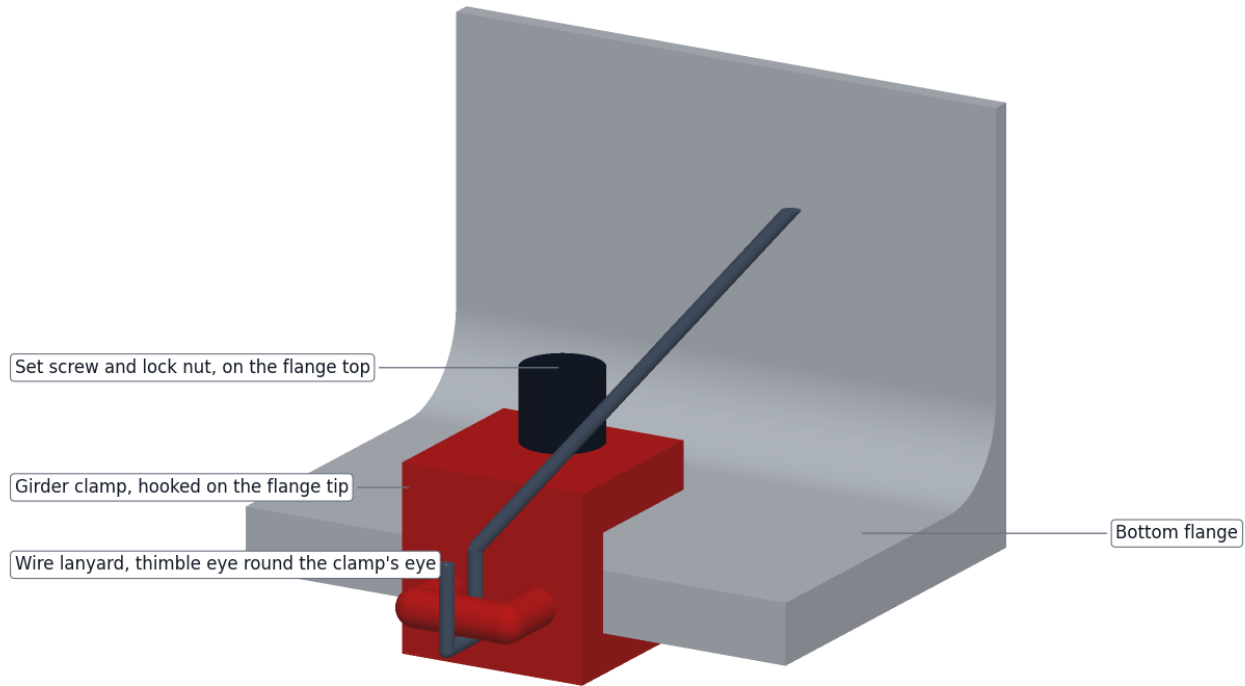
How it fits. The back plate bears flat on the post's outer face, with the box base 450 above the deck. Each band goes round the post, along the back of the plate to a slot, through it, across the plate's front and back through the other slot; its worm-drive housing sits behind the post.

Check before moving on. Each band closes on the post with adjustment to spare, and the node does not turn on the post when pushed by hand.

3.11 Lanyard and its girder clamp (bought)

Joint 10: lanyard girder clamp on the bottom flange

Hooked on the flange tip 159 mm along the span from the nearer foot clamp; nothing is drilled



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github.com/BoujeeEnjinia1701/bridgepulse

Figure 22. The girder clamp hooked on the edge of the bottom flange, its set screw on the flange top and the lanyard's eye round the clamp's eye.

What it is. The second hold on the bridge, in case the foot clamps or the jack ever let go. Three bought parts: a stainless pad eye with a 38 x 22 base and two M5 x 10 screws; a 3 mm stainless wire lanyard about 0.45 m long with a thimble eye and a crimped ferrule at each end; and a single-flange girder clamp with an eye, a set screw and a lock nut, whose maker rates it for a working load of 100 kg or more.

How it fits. The pad eye is screwed to the plate's front at the top left, above the hub, into the two tapped holes of section 3.1, its screws ending 1.5 short of the web. The girder clamp hooks over the edge of the bottom flange with its lower jaw under the flange, 150 or more along the span from the nearer foot clamp (159 on the example) on the left, and its set screw is tightened down onto the flange top and locked. One end of the lanyard goes round the pad eye's loop and the other round the clamp's eye. The lanyard runs straight past the left end of the hub with no more than 25 of slack, since the slack sets how hard the lanyard is snatched if the mount lets go.

Check before moving on. The clamp cannot be pulled off the flange by hand; the lanyard clears the hub and every cable; the slack is 25 or less.

3.12 Bought components

Buy to specification, not brand. Line numbers are those of the bill of materials.

- **Hub enclosure (line 1).** Die-cast aluminium box, IP67, 170 x 64 x 110 outside, gasketed lid; two M16 x 1.5 nylon glands for 4 to 8 cable and two M12 x 1.5 glands for 3 to 6.5 cable, each IP68.
- **Accelerometer board (line 2).** ADXL355 class, 22.5 µg/VHz, SPI, on a breakout with four mounting holes; four 5 metal spacers.
- **Signal board (line 3).** As section 3.6; four 14 standoffs; bridge completion resistors 10 k, 0.1 %.
- **Mounting fixings and lanyard (line 4).** Stainless: one M12 x 70 hex screw and one M12 nut; two M10 x 35 hex bolts and washers; six M6 x 30 cap screws; four M5 x 12 screws with bonded sealing washers; nylon isolating washers. Lanyard: as section 3.11.
- **Strain gauges (line 5).** Four 350 Ω foil gauges self-temperature-compensated for steel, from one batch; cyanoacrylate gauge adhesive; surface preparation kit; gauge coating and four cover patches about 60 x 40; neutral-cure silicone.
- **Temperature probes (line 6).** Two TMP1826 class sensors on carriers, two stainless sheaths 7 x 60, sealed 3-core lead, potting epoxy, adhesive-lined heat-shrink.
- **Cables and clips (line 7).** As section 3.9, and the M12 5-pin panel connector (rear mounting, M16 thread).
- **FieldNode core (line 8).** As section 3.10.
- **microSD card (line 9).** 32 GB, rated -25 to +85 °C.
- **Installation banding (line 10, installed units only).** Two lengths of 12.7 x 0.76 stainless banding, about 0.4 m each, with two one-way crimped buckles, which replace the FieldNode's worm-drive band clamps on a bridge open to the public. The bench unit keeps the worm-drive bands. Fitting it needs a banding tool.
- **Bench rig (not in the bill of materials).** Two offcuts of the girder section about 650 long (IPE 360 on the example bridge), long enough for the lanyard's girder clamp, set up on the bench 1.2 m apart, centre to centre; a 1.5 m stub of 50 x 50 x 3 square hollow section held upright in a stand.

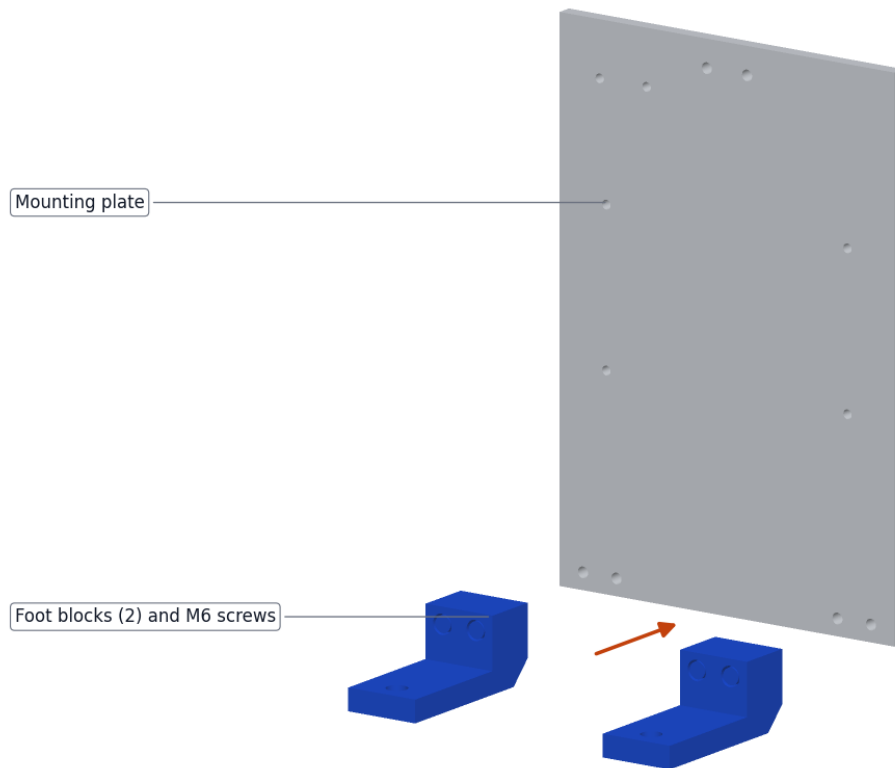
■ 4. Putting it together

In each picture the parts already fitted are grey and the part being fitted is in colour, with an arrow showing the way it goes in.

Step 1: foot blocks onto the plate

Step 1: foot blocks onto the plate

Back face of each block on the plate's front, chamfer down; two M6 x 30 cap screws each



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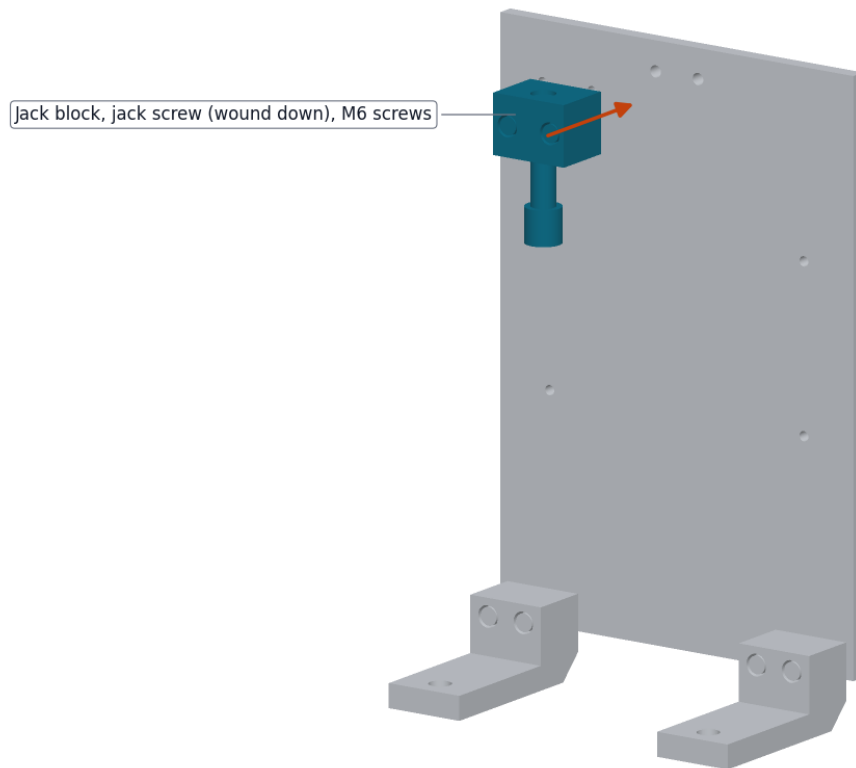
github.com/BoujeeEnjinia1701/bridgepulse

Lay the plate front up. Put each foot block's back face on the plate's front at the bottom, chamfer down, and fit two M6 x 30 cap screws, tight.

Step 2: jack block and jack screw onto the plate

Step 2: jack block and jack screw onto the plate

Top flush with the plate's top edge; two M6 x 30 screws. Wind the jack screw down below the block top



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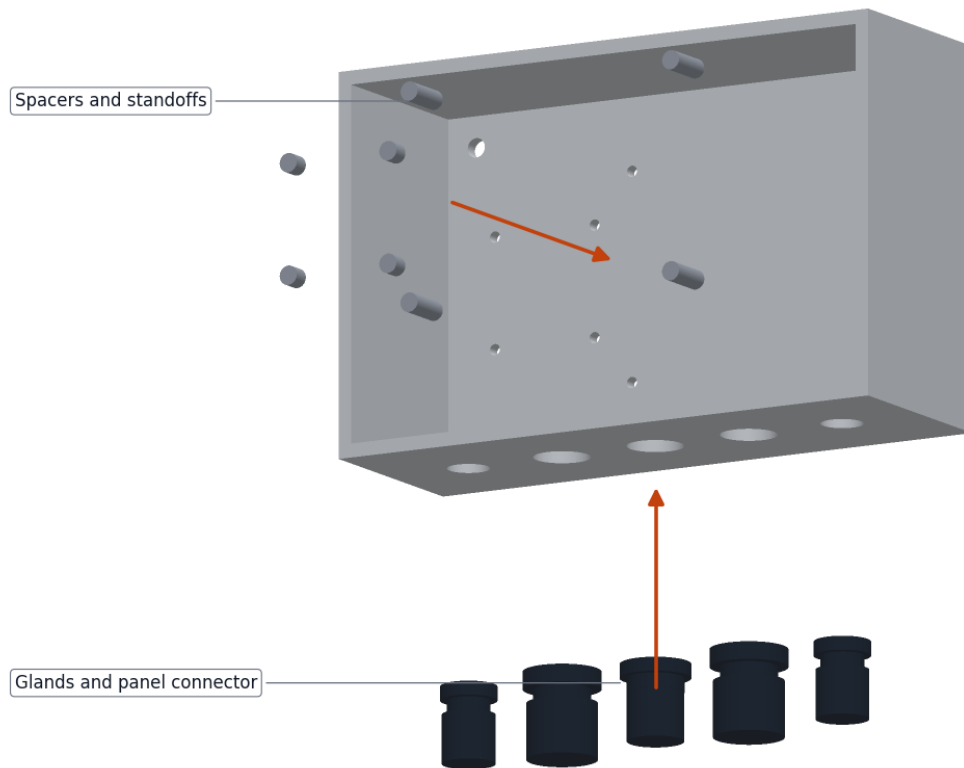
github.com/BoujeeEnjinia1701/bridgepulse

Fit the jack block at the top of the plate's front, its top flush with the plate's top edge, with two M6 x 30 cap screws. Run the lock nut onto the M12 screw, then screw it up into the block from below until its end is just below the block's top.

Step 3: glands, connector, spacers and standoffs into the hub

Step 3: glands, connector, spacers and standoffs into the hub

Penetrations from below, seal outside, nut inside. Spacers inside on M3 countersunk screws from outside the base



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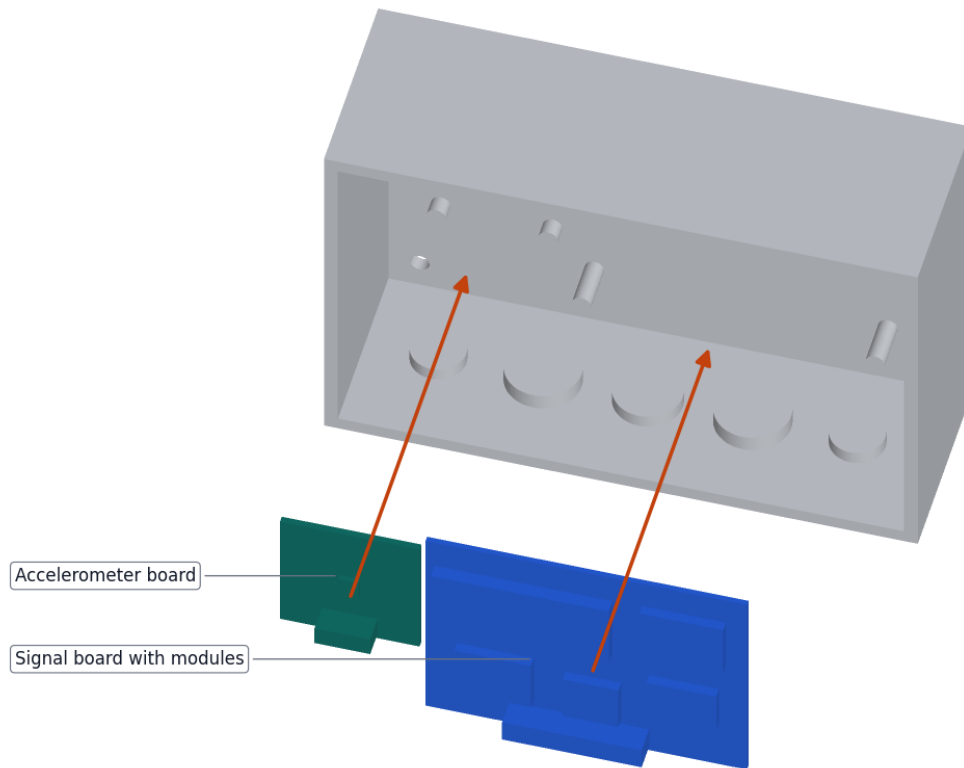
github.com/BoujeeEnjinia1701/bridgepulse

As section 3.5, steps 5 and 6. Fit blanking plugs in any gland whose cable is not yet ready.

Step 4: boards into the hub, then wire them

Step 4: boards into the hub, then wire them

Accelerometer board on its four spacers, signal board on its standoffs; wire as the wiring diagram shows



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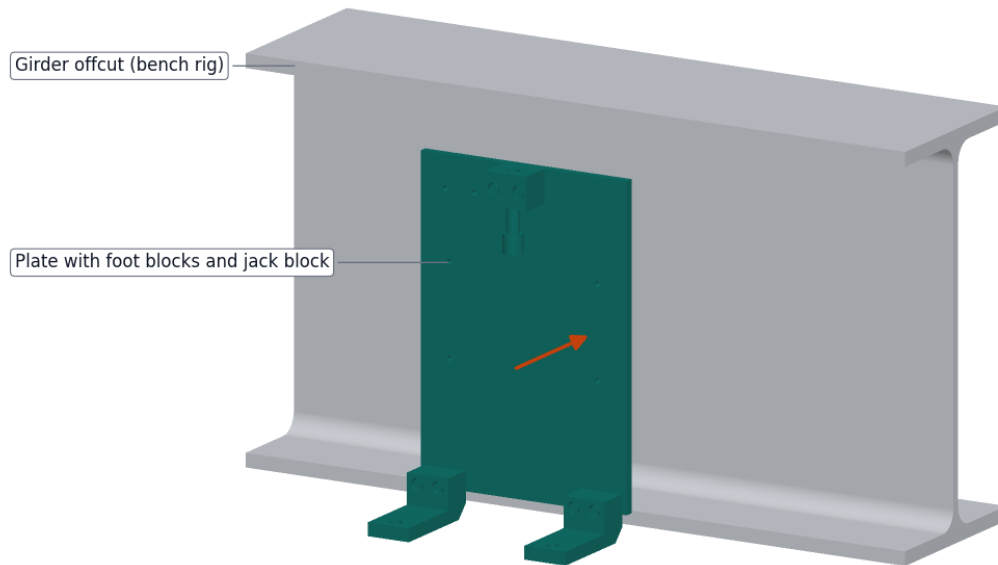
github.com/BoujeeEnjinia1701/bridgepulse

Fit the accelerometer board and the signal board, and wire them as Figure 15. **Hold point:** the wiring checks of section 3.6 pass.

Step 5: plate onto the girder

Step 5: plate onto the girder

Back of the plate flat on the web, foot blocks standing on the bottom flange, midway along the span



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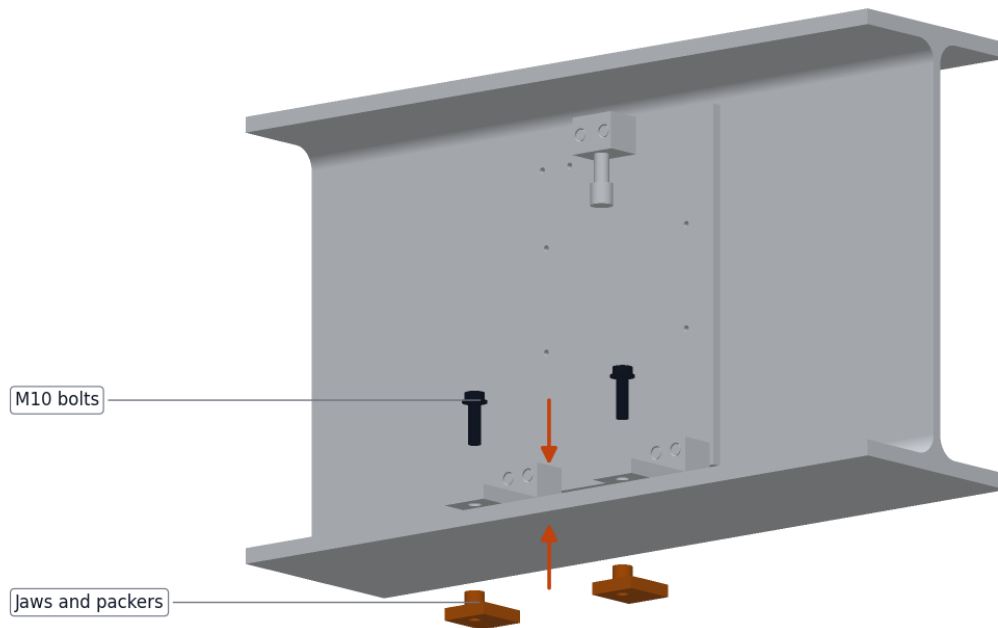
github.com/BoujeeEnjinia1701/bridgepulse

Hold the plate with its back flat on the outer face of the south girder's web, its foot blocks standing on the bottom flange, midway along the offcut.

Step 6: clamp the foot blocks to the flange

Step 6: clamp the foot blocks to the flange

Jaw under the flange, packer on the jaw beside the tip, bolt down through tail and packer; snug only



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

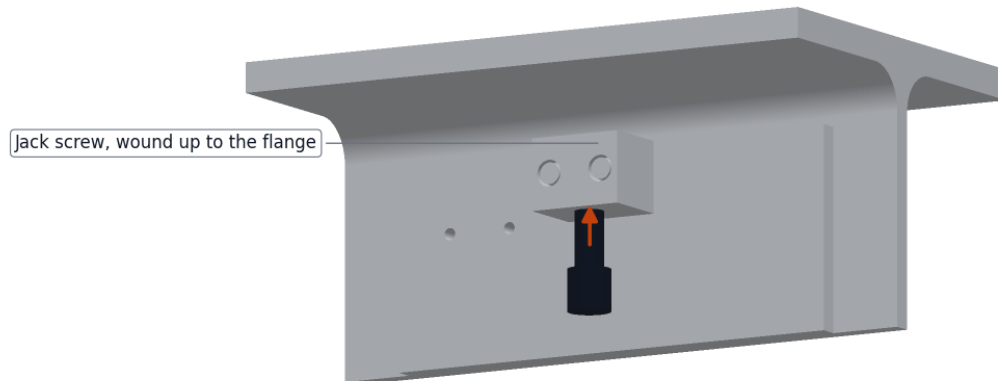
github.com/BoujeeEnjinia1701/bridgepulse

For each block, slide a jaw under the flange with its tapped end out past the flange edge, stand a packer on the jaw beside the edge, and fit an M10 x 35 bolt and washer down through the tail and the packer into the jaw. Snug only.

Step 7: jack screw up to the top flange

Step 7: jack screw up to the top flange

Wind the screw up until it bears, then a further quarter turn; tighten the lock nut, then the clamp bolts



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

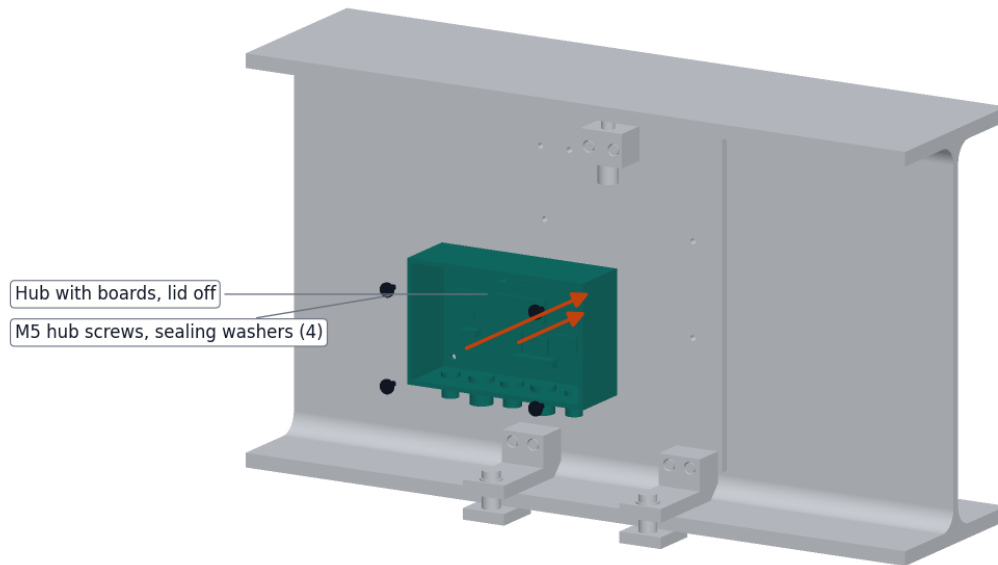
github.com/BoujeeEnjinia1701/bridgepulse

Wind the jack screw up until its end bears on the top flange, then a further quarter turn. Tighten the lock nut. Then tighten both clamp bolts to the torque the bolt maker gives for their grade, and record it.

Step 8: hub onto the plate

Step 8: hub onto the plate

Base flat on the plate; four M5 x 12 screws from inside the box, sealing washer under each head



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

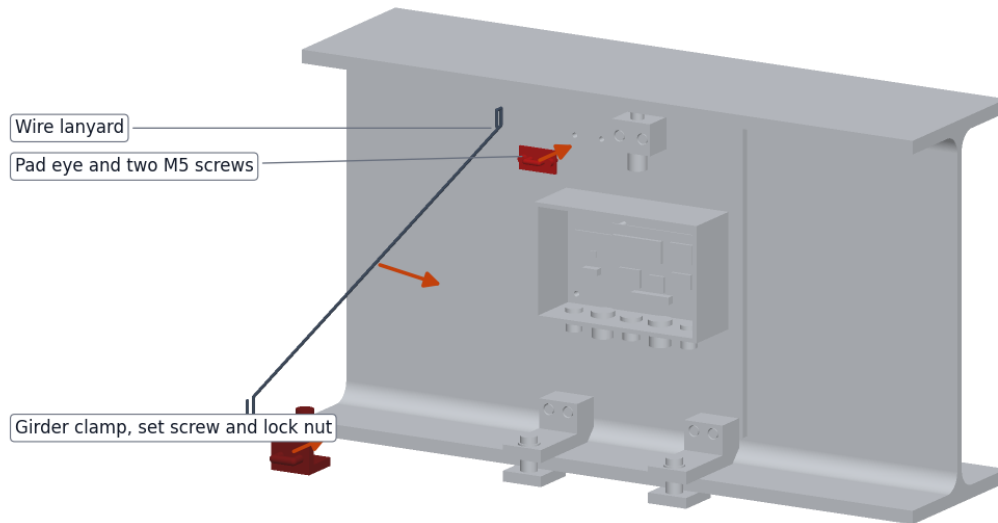
github.com/BoujeeEnjinia1701/bridgepulse

With the lid off, hold the hub's base flat on the plate, centred, and fit the four M5 x 12 screws from inside the box, a bonded sealing washer under each head. Tighten evenly.

Step 9: lanyard and its girder clamp

Step 9: lanyard and its girder clamp

Pad eye on the plate above the hub; clamp hooked on the flange tip 150 mm or more from the foot clamps; lanyard between them



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

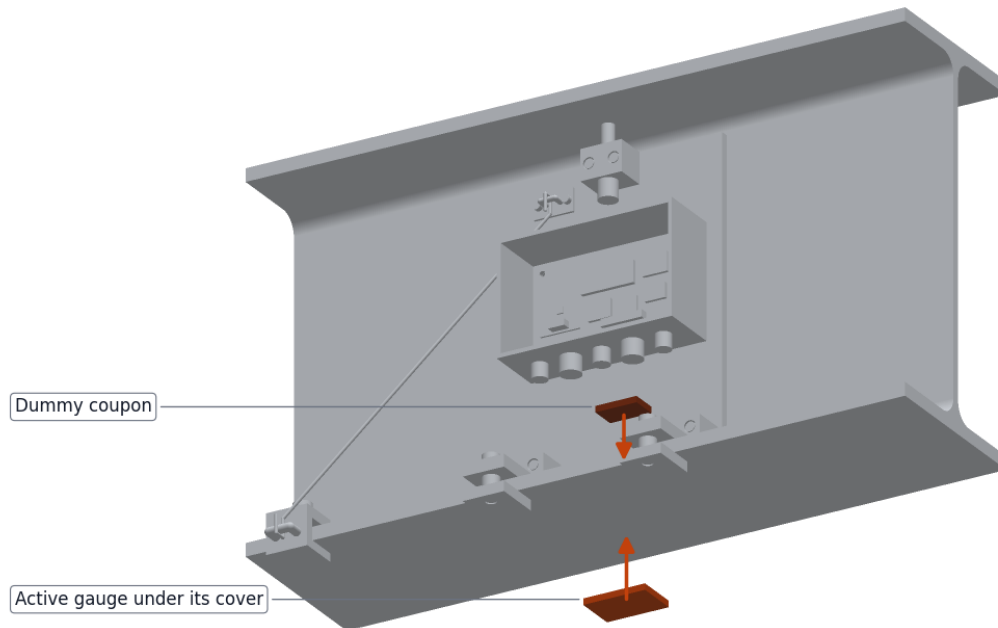
github.com/BoujeeEnjinia1701/bridgepulse

Screw the pad eye to the plate's front at the top left with its two M5 x 10 screws. Hook the girder clamp over the edge of the bottom flange 150 or more to the left of the left foot block, lower jaw under the flange, and tighten its set screw onto the flange top; lock it with its nut. Fit the lanyard between the pad eye and the clamp's eye as section 3.11, with no more than 25 of slack. **Hold point:** the lanyard is fitted before anyone works below the hub.

Step 10: strain gauge and dummy coupon

Step 10: strain gauge and dummy coupon (south girder; the north girder is the same)

Paint removed only with the owner's permission; gauge bonded under the flange; coupon on silicone



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

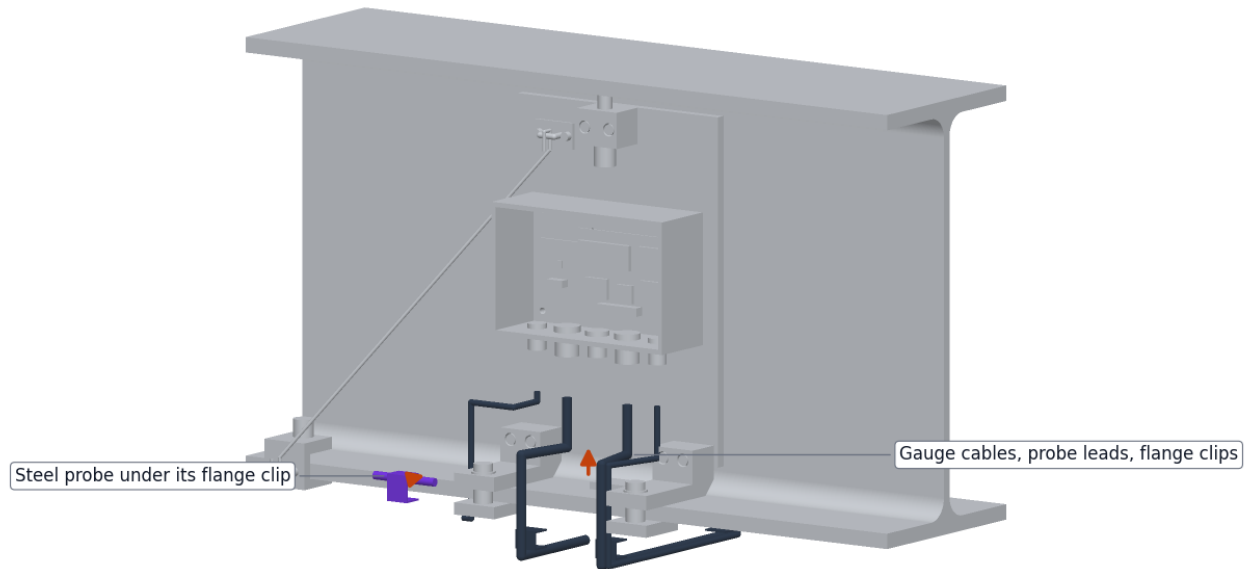
github.com/BoujeeEnjinia1701/bridgepulse

On each girder offcut: prepare a patch of bare steel under the bottom flange on the girder's centre line, bond the active gauge along the girder, solder its leads, coat it and fit its cover. Set the dummy coupon on silicone on the flange top as section 3.7. On a real bridge, paint is removed only with the owner's permission and after a lead test (section 6, S4).

Step 11: steel probe, cables and clips

Step 11: steel probe, cables and clips

Probe under its clip on the flange; cables tied to flange clips, into their glands from below; glands tightened



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

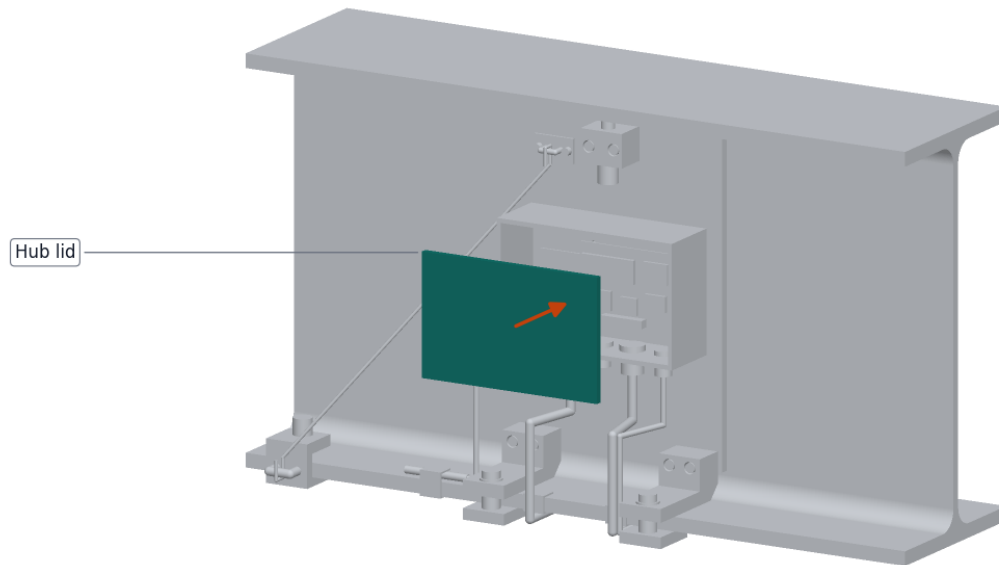
github.com/BoujeeEnjinia1701/bridgepulse

Push the probe clip onto the flange edge over the steel probe, with thermal paste under the probe. Push the cable clips onto the flange edges, run the near gauge cable and the steel probe lead to their glands, tie them to the clips, and tighten the glands on them. Connect them at the terminal block.

Step 12: close the lid

Step 12: close the lid

Fresh desiccant pack inside; gasket clean, no wire across it; lid screws tightened evenly in a cross pattern



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

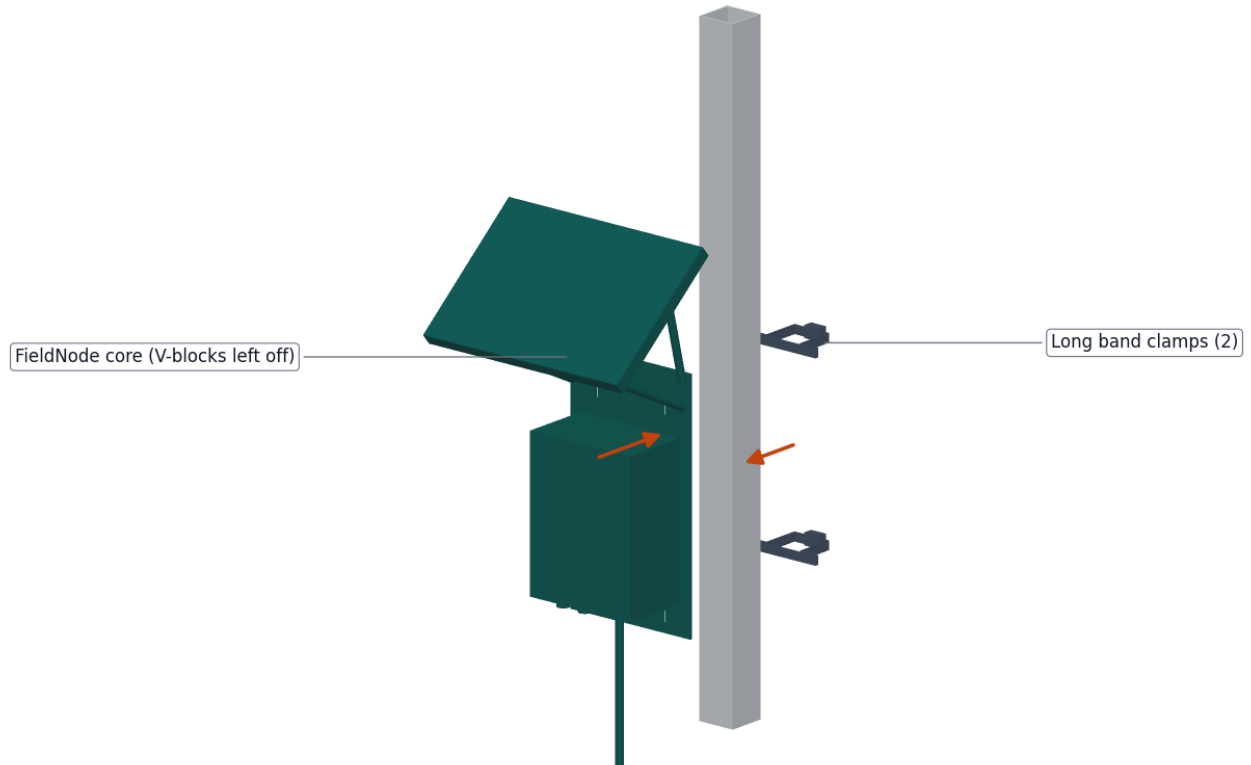
github.com/BoujeeEnjinia1701/bridgepulse

Put a fresh desiccant pack in the hub. Check the gasket is clean and seated with no wire across it, and tighten the lid screws evenly in a cross pattern.

Step 13: FieldNode core onto the post

Step 13: FieldNode core onto the post

Back plate flat on the post's face; each band round the post, through both slots and across the plate front



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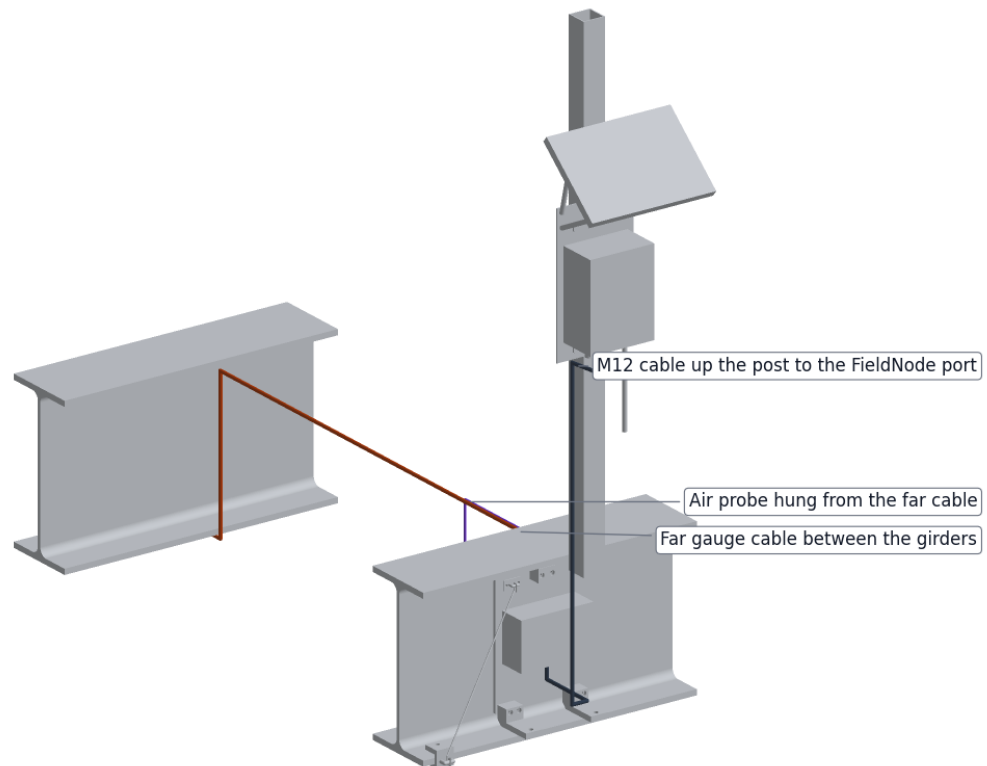
Hold the FieldNode's back plate flat on the post's outer face, its box base 450 above the deck level. Pass each band round the post, through both slots and across the plate's front, worm-drive housing behind the post, and tighten.

Hold point: the FieldNode core has passed its own plan's safety stops before its cell is fitted.

Step 14: cable to the FieldNode, far gauge cable and air probe

Step 14: cable to the FieldNode, far gauge cable and air probe

M12 cable clipped up the post to the FieldNode port; far cable across between the girders, air probe hung from it



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github.com/BoujeeEnjinia1701/bridgepulse

Plug the M12 cable into the hub's panel connector, run it out past the flange edge and up beside the post, clipped to it, and into the FieldNode's port B. Run the far gauge cable from the second girder's gauge under its flange, across between the girders and under the south flange to the right gauge gland. Hang the air probe on its lead from the far cable's crossing, about 110 below it, tie its lead along the far cable to the right probe gland, and tighten both glands. On the bench, tie the crossing to the bench frame.

■ 5. First checks

These are the checks a TRL 4 test report would record; this plan only lists them. Requirement numbers are those of BRP-REQ-001.

Table 2. First checks.

Check	Requirement	How	Pass when
Penetrations and lid sealed	R10	Look at each seal under a lamp; gland caps tight on their cables	Every seal evenly squeezed; no gap at the lid gasket
Lanyard fitted	Safety	Look along the lanyard; pull the girder clamp by hand	Clamp 150 mm or more from the foot clamps and firm; slack 25 mm or less; nothing rubs on the lanyard
Mount is firm	R1, R2	Push the hub by hand in every direction; tap the hub with a soft mallet while recording the accelerometer	Nothing moves at any joint; the hub's own mode in the record is above 120 Hz
Accelerometer noise	R1	Record 10 minutes on a quiet bench at night	Noise density 25 $\mu\text{g}/\text{VHz}$ or less in the 0.5 to 60 Hz band
Strain channels	R4	Read both channels at rest, then put a known shunt resistor across each active gauge	Steady readings; the shunt step matches its calculated strain within 2 μe
Temperature probes	R5	Ice-point check (section 3.8), then both probes beside a reference in still air	0 °C within 0.5 °C; within 0.5 °C of the reference
Power at the port	R6	Bench supply at 5 V with a 200 mA limit in place of the FieldNode port; measure current through a 10 minute record	About 19 mA while recording (93 mW); 0 when the rail is off
Summary and storage	R7	Run one record; read the summary bytes on the RS-485 link and the raw file on the card	36 bytes, or 11 in reduced mode; raw file about 1.9 MB
No drilling, fitting time	R8	Time the plate, hub and clamps going on the offcut	Nothing drilled into the steel; time recorded
Depth below the girder	R9	Straight edge across the underside of the bottom flange; measure the jaws, covers, clips and cables	15 mm or less everywhere (10 mm by the model)
Hub inside the girder outline	R9	Straight edge down from the flange edge	The closed hub's front face is inside the edge line

■ 6. Safety stops

Stop at each point. Carry on only when everything listed is true.

- **S1. Before the girder offcuts are moved.** Two people to lift each; the offcuts stand on their bottom flanges on a firm floor or low trestles that cannot tip; feet and hands clear.
- **S2. Before any steel from a bridge is ground or sanded.** A lead test of its paint is negative, or the paint is removed by a trained contractor under local abatement rules.
- **S3. Before gauge adhesive, conditioner or potting epoxy is opened.** Safety data sheets read; gloves and eye protection on; the room ventilated; no food nearby.
- **S4. Before paint is removed from a real bridge.** The owner's written permission is in hand, and the lead test of S2 is done on that bridge.
- **S5. Before the hub is first powered.** The bench supply is set to 5 V with a 200 mA current limit; the panel connector's pin assignment is checked against the wiring with a meter, not by wire colour.

- **S6. Before the FieldNode core's cell is fitted.** Its own plan's safety stops are passed (FND-BLD-001).
- **S7. Before the rig is left unattended.** The jack lock nut and both clamp bolts are tight and their torques recorded; the lanyard is fitted; no cable is under strain.
- **S8. Before any installation on a bridge (outside this plan).** The owner's written permission; a trained crew of at least two; fall protection and a rescue plan for work over water; traffic management where needed; the lanyard fitted to its anchor, a second girder clamp rated for 100 kg or more working load on the bottom flange at least 150 mm along the span from the foot clamps, with no more than 25 mm of slack; the FieldNode's bands replaced by stainless banding with one-way crimped buckles. Never work alone.

■ 7. Tools, skills and workspace

Tools. Hacksaw with a 24 teeth per inch blade, or a bandsaw; bench vice with soft jaws; bench drill or a drill in a stand; drills 3 to 12; step drill to 20; 11 counterbore (or an 11 flat-bottomed drill); countersink; taps M5, M6, M10 and M12 with their tap drills (4.2, 5.0, 8.5, 10.2); flat and half-round files; deburring tool; scribe, engineer's square, 45° square, steel rule and calipers; torque wrench covering about 5 to 40 N·m; spanners and hex keys; soldering iron; ferrule crimper and wire strippers; multimeter; bench power supply with an adjustable current limit; strain gauge installation kit; heat gun; ice bath and a reference thermometer; lead paint test kit; stopwatch.

Skills. No certified trade is needed. Basic metalwork (marking out, sawing, drilling, tapping, filing), through-hole soldering and crimping, and strain gauge bonding, which is best practised first on a scrap of steel. All circuits are extra-low voltage: 5 V at the port and 3.3 V inside the hub. The bench supply must be a certified, undamaged unit; no mains wiring is part of this build.

Workspace. A bench about 1.5 x 0.6 m with room beside it for the two girder offcuts 1.2 m apart; a metalwork corner kept apart from the electronics; a clean, dust-free, ventilated place for bonding gauges and potting probes.

Personal protective equipment. Safety glasses for cutting, drilling, soldering and chemicals; nitrile gloves for adhesives and epoxy; cut-resistant gloves for bar and plate; safety boots for the offcuts; hearing protection when sawing; no gloves near a turning drill.

■ 8. Where the numbers come from

- Model and constructability checks: `cad/src/model.py` (`python cad/src/model.py --check` , 91 checks); STEP and STL exports in `cad/step/` and `cad/stl/` .
- Pictures: `cad/src/build_plan_media.py` , using `.kit/build_views.py` ; written to `docs/05-build-plan/` and `cad/drawings/BRP-DWG-101` to `BRP-DWG-107` .
- General arrangement: `cad/drawings/BRP-DWG-001.pdf` , Rev P6.
- Calculations: `docs/04-calcs/01-sizing.md` (BRP-CAL-001 v0.6) and `docs/04-calcs/sizing.py` ; mass [H1], mount stiffness [H2], retention [H4], lanyard [H4a], [H4c], root fillets [H4b], depth below the girder [H5], FieldNode on the post [H7], [H8], power [F2], [F3], cost [K1].
- Bill of materials: `bom/bom.csv` and `bom/bom-notes.md` .
- Decisions: `docs/decisions/0004-design-for-construction.md` (BRP-DDR-004), with BRP-DDR-001 to BRP-DDR-003; decisions and items to confirm in `docs/06-design-decisions.md` (BRP-DEC-001).
- Requirements: `docs/03-requirements.md` (BRP-REQ-001 v0.8).
- FieldNode core: FieldNode's build plan FND-BLD-001, in the FieldNode repository.